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Ionosphere Variations

Transport

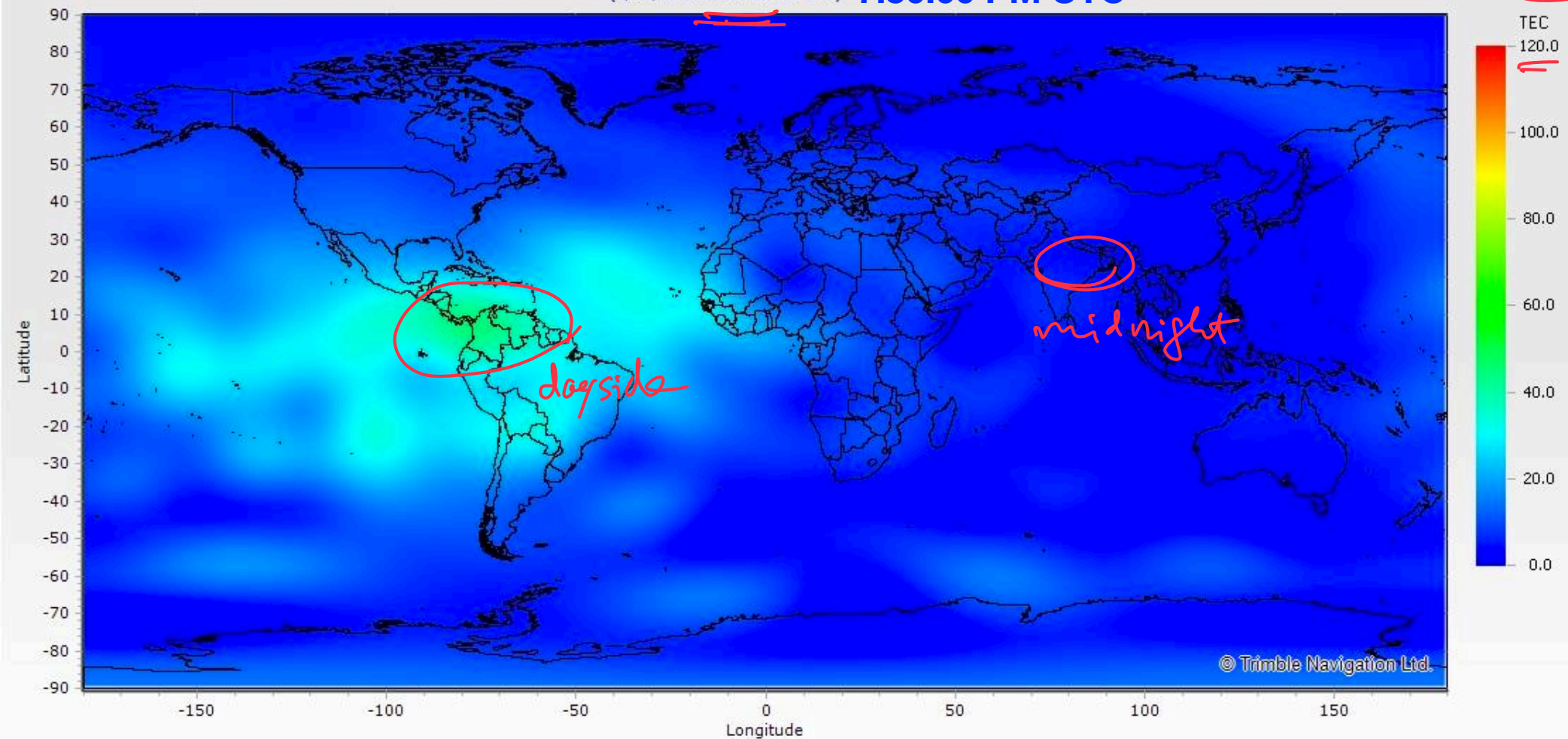
- ❖ Variety of processes move plasma in the ionosphere:
 - ❖ **Winds**: neutral winds drag ions along with them, if collision frequency is high enough
 - ❖ **Drifts**: various forces cause plasma to “drift”: electric and magnetic fields; gravity; pressure; etc.
- ❖ **These contribute to complex spatial and temporal variations in the ionosphere**

Global variation: Snapshots

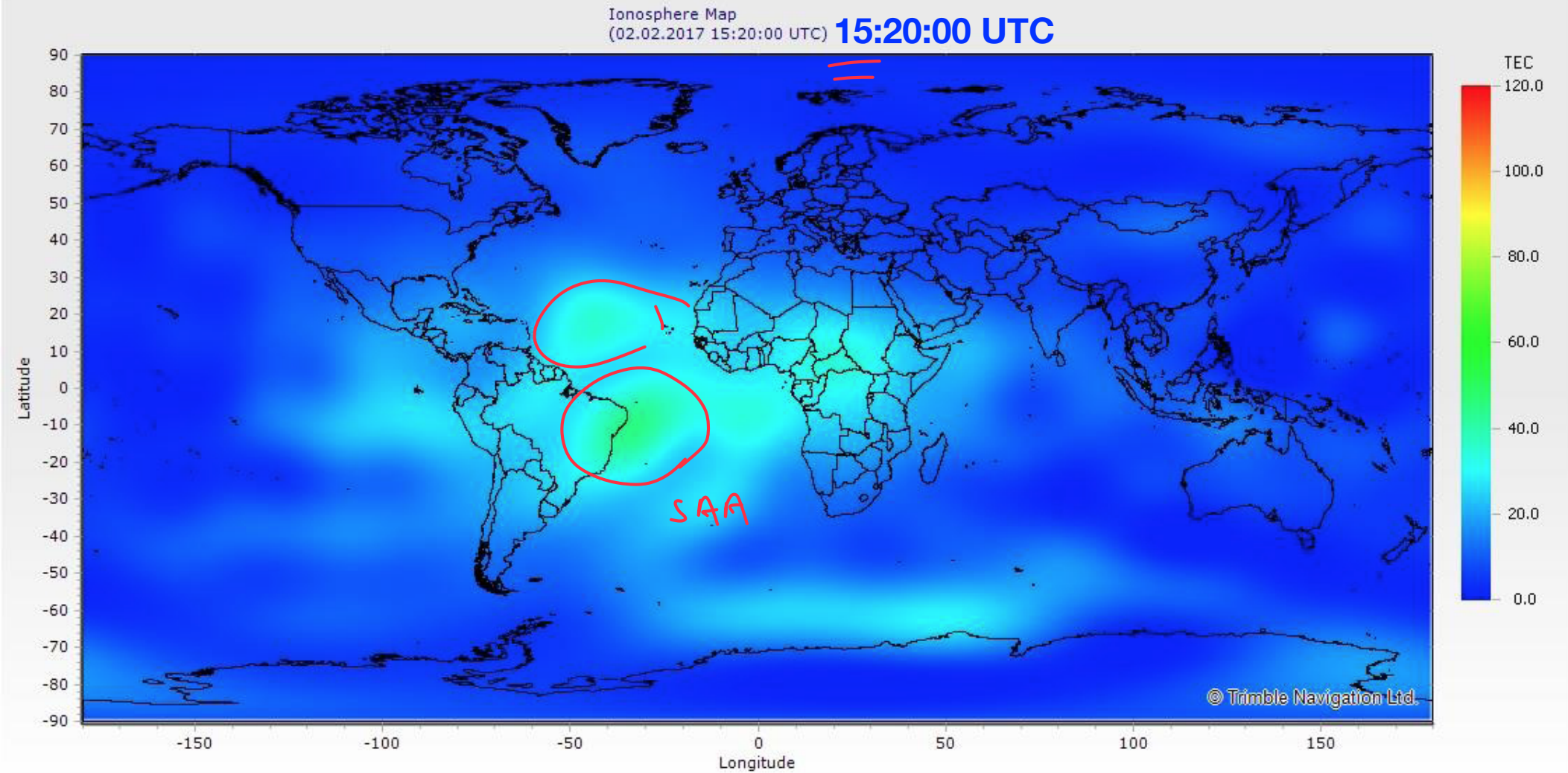
$$TEC = \int_0^{\infty} n_e(z) dz$$

$$1 \text{ TECU} = \frac{10^{16}}{\text{cm}^2}$$

Ionosphere Map
(9/26/2018 7:50:00 PM UTC) **7:50:00 PM UTC**



Global variation: Snapshots



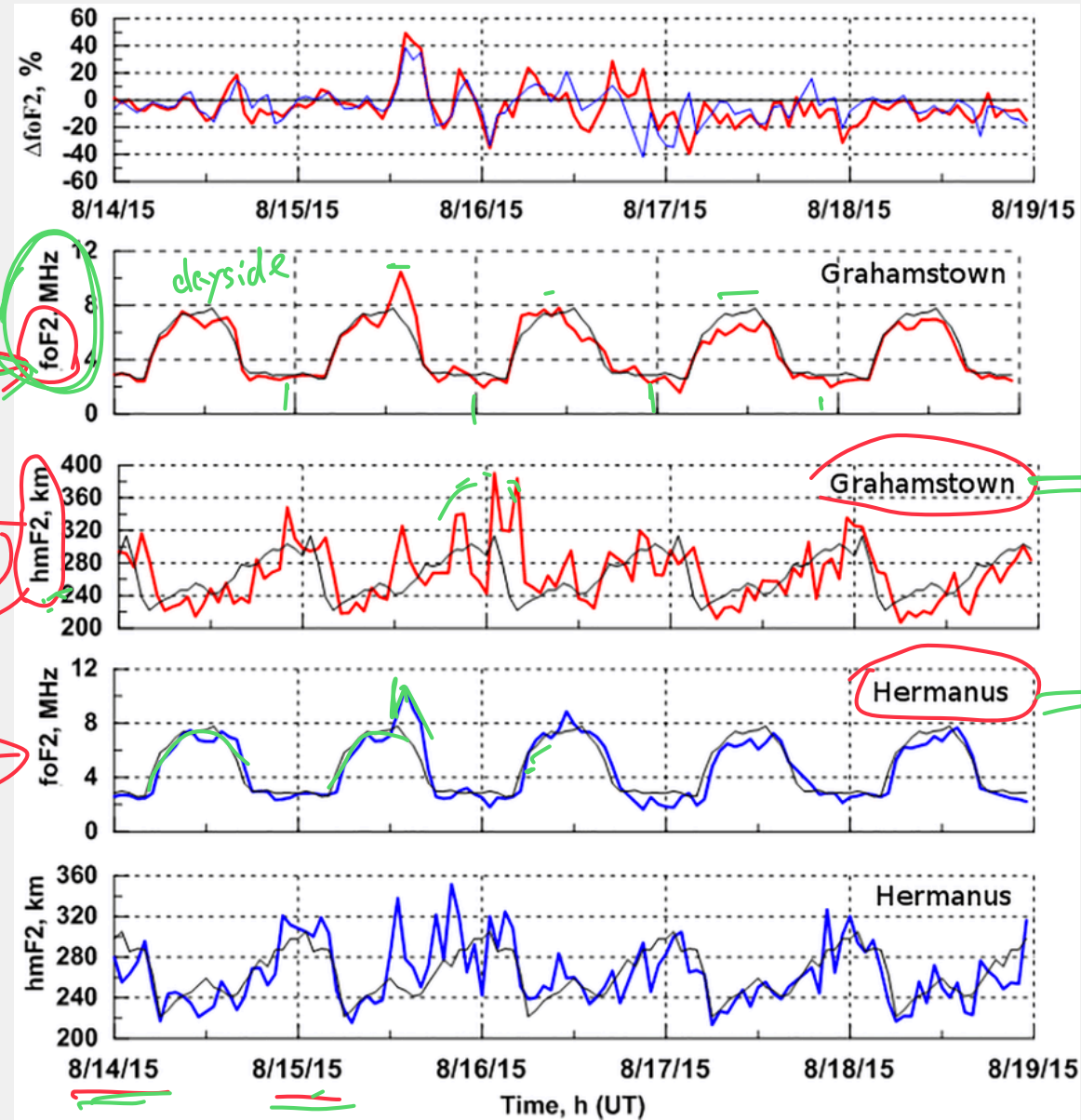
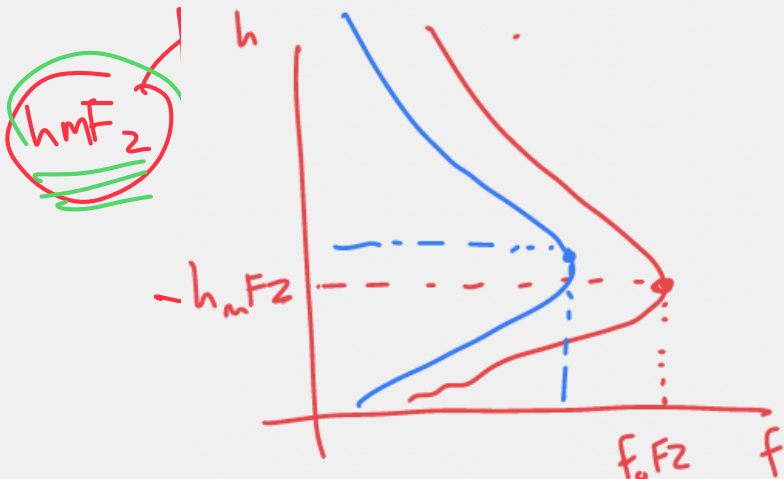
foF2 and hmF2

- ❖ We often characterize the ionosphere by its peak density, foF2, and the altitude where that occurs, hmF2
- ❖ foF2 is a frequency in MHz. related to electron density:

$$\omega_{pe} = \frac{\sqrt{n_e e^2}}{m_e \epsilon_0} \sim \sqrt{n_e}$$

$$f_{pe} \text{ (MHz)} = \frac{\omega_{pe}}{2\pi} = 9 \sqrt{n_e (\text{cm}^{-3})}$$

- ❖ hmF2 is simply altitude in km

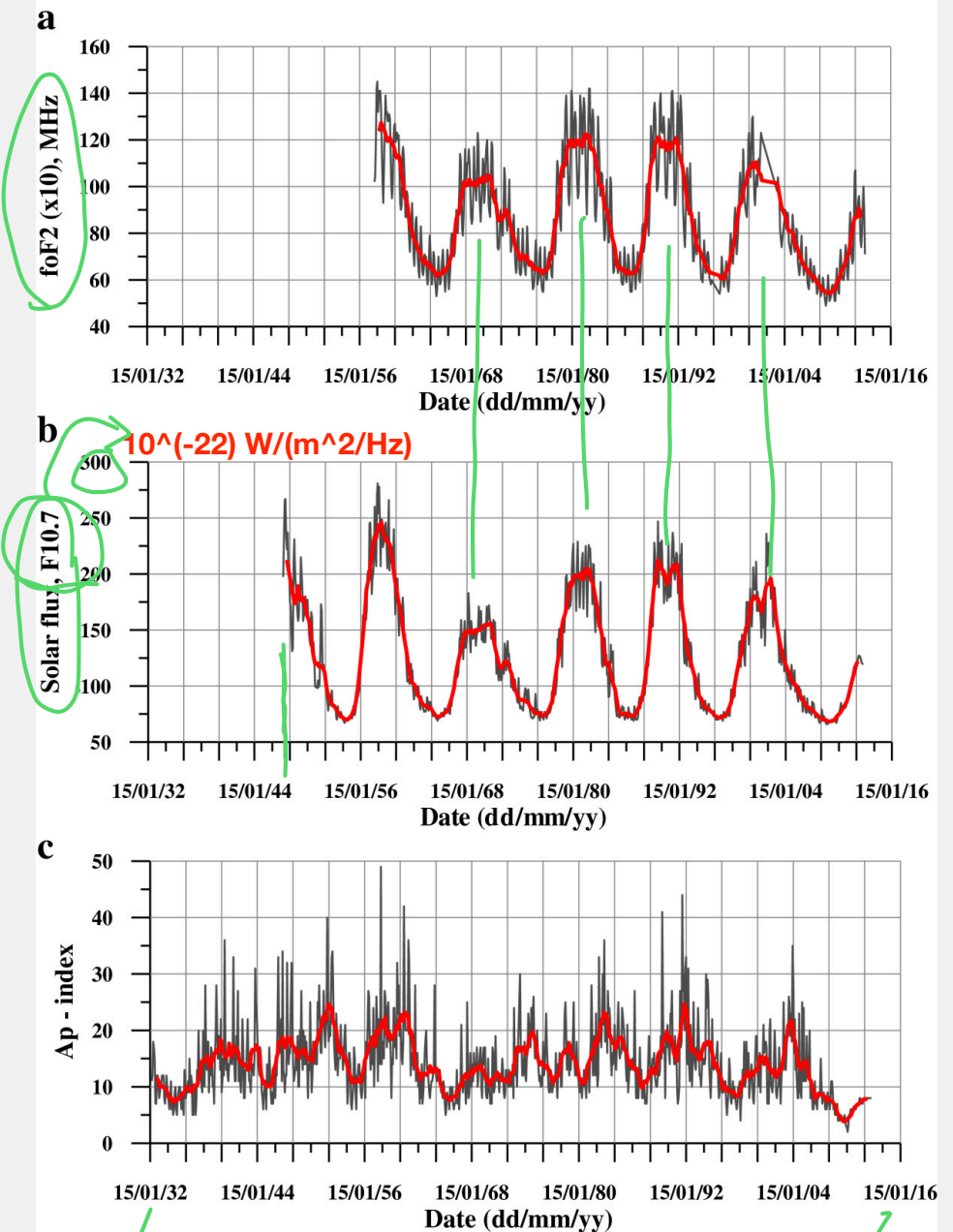
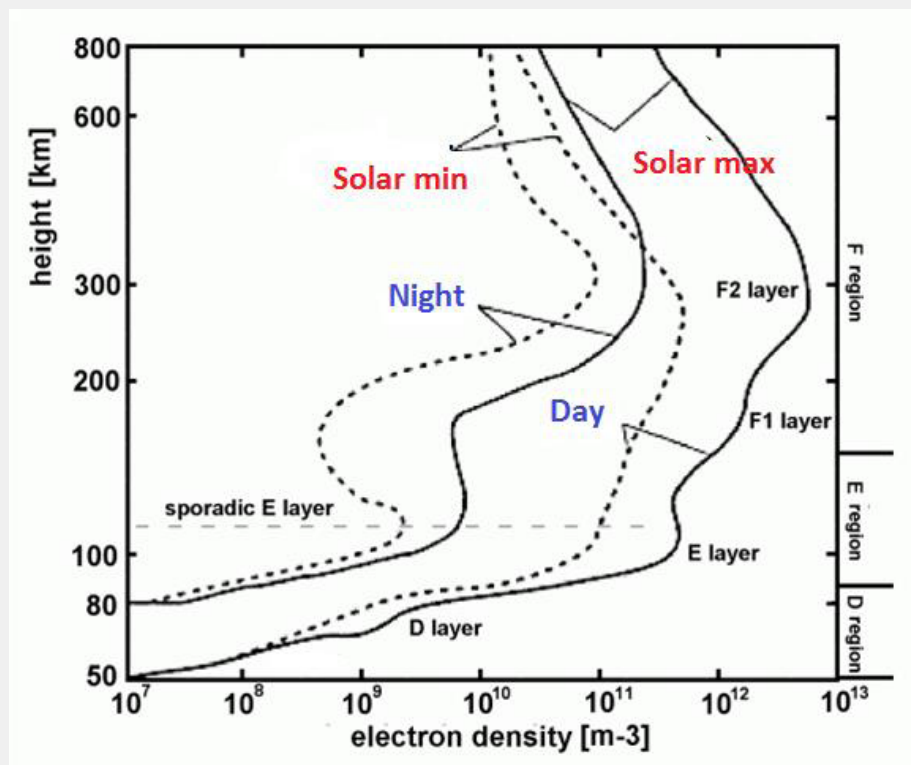


How to determine foF2 and hmF2 ?

Ionosonde

Solar Cycle Variation

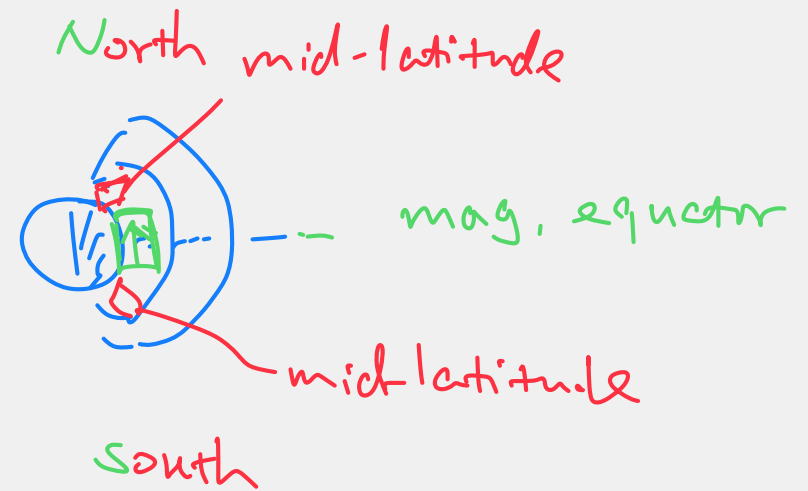
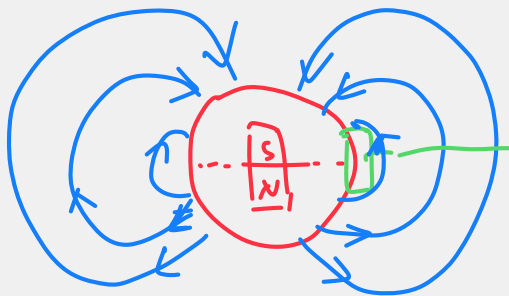
- ❖ Densities are much higher (order of magnitude) at solar maximum compared to solar minimum
- ❖ Higher EUV / X-ray fluxes lead to higher ionization rates



1932

2016

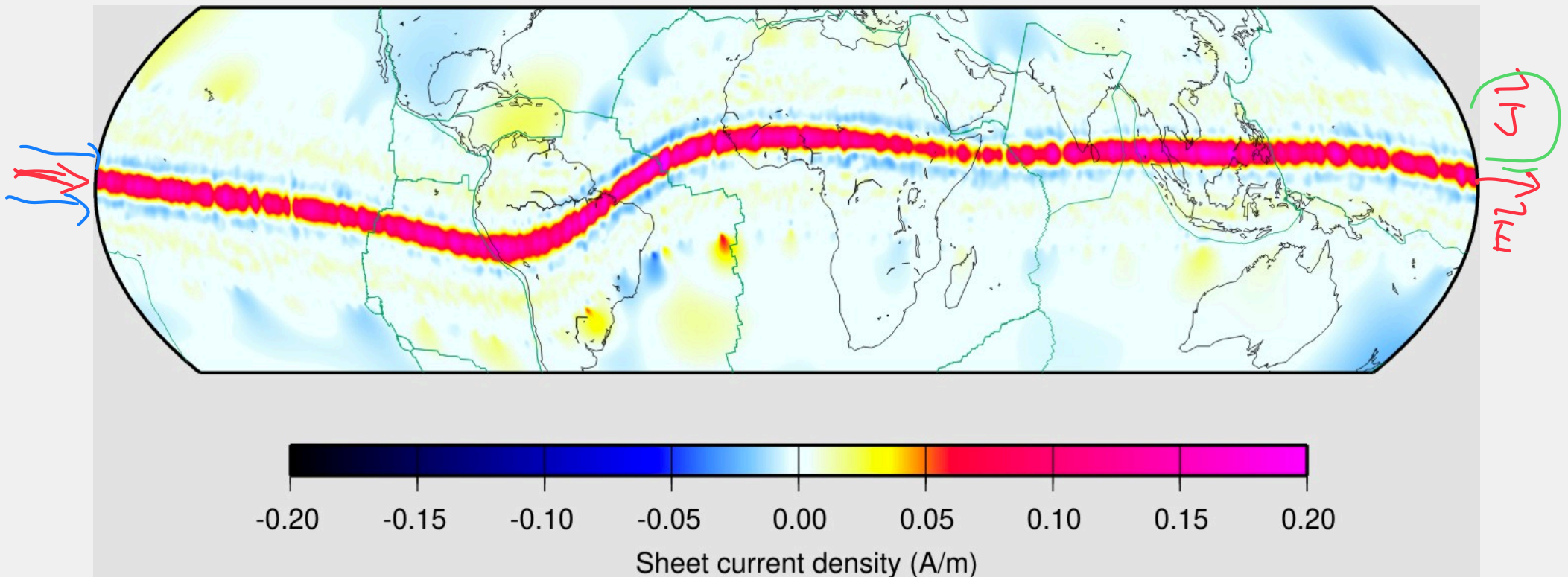
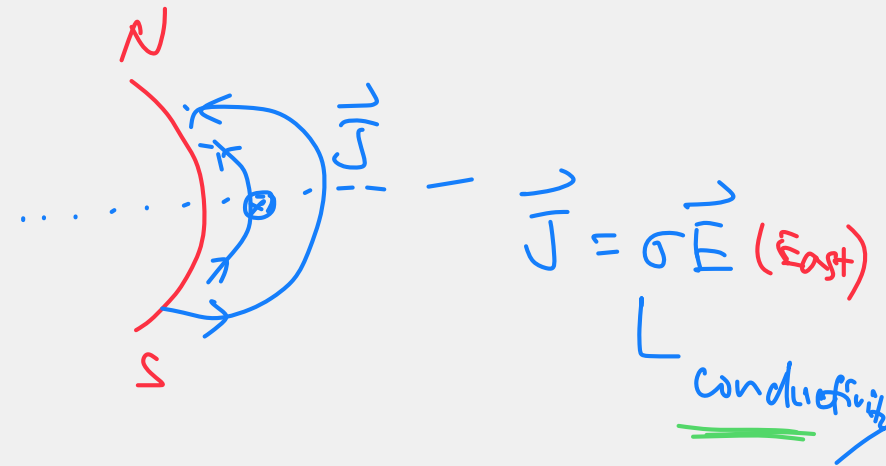
Magnetic field



Equatorial Electrojet (EEJ)

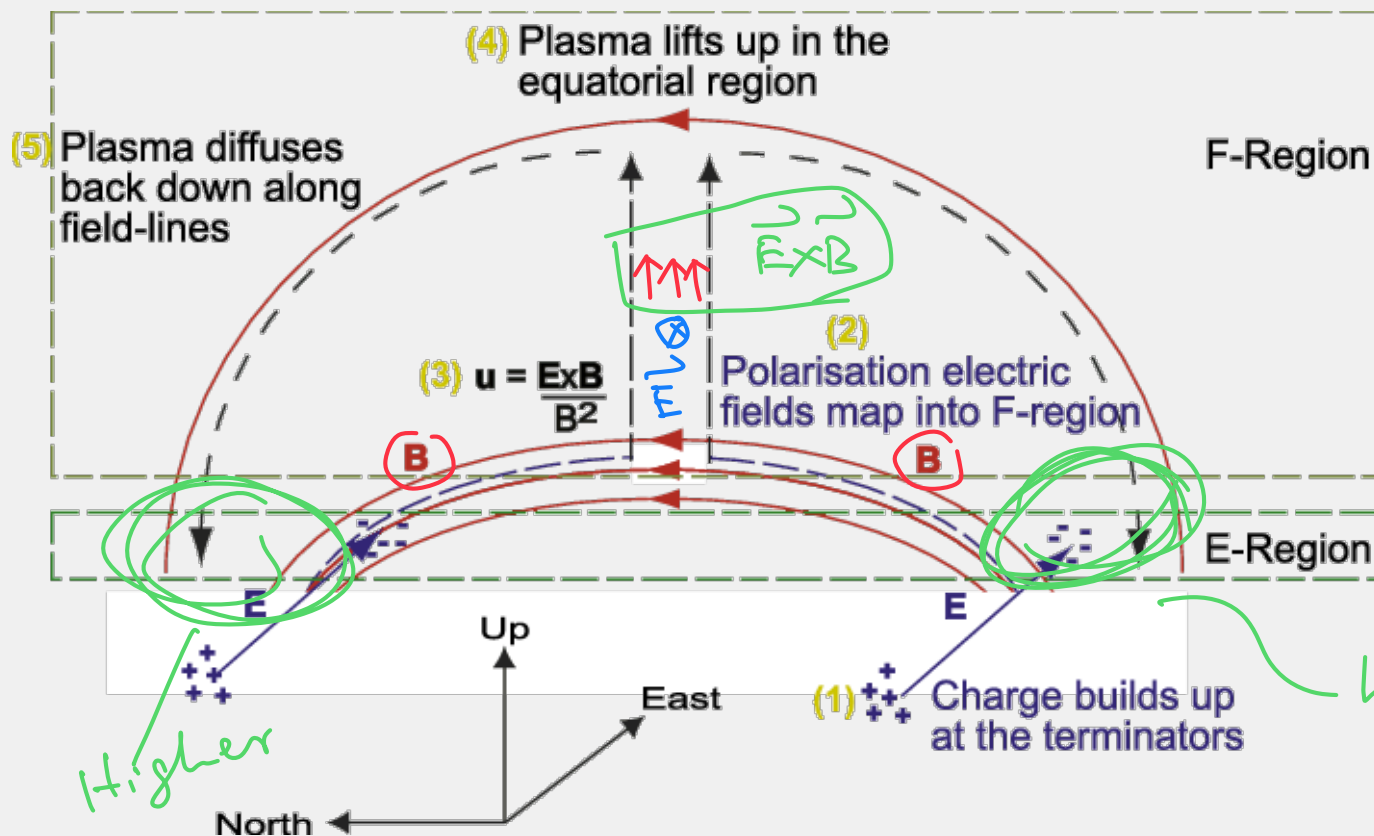
- ❖ Plasma physics involving B-field lead to an intense **current** that flows East in the dayside ionosphere
- ❖ Restricted to narrow region in latitude; 110-130 km altitude

E-region



Equatorial (ionization) Anomaly

- ❖ Due to the equatorial electrojet (EEJ) current, an electric field arises
- ❖ $E \times B$ leads to a **drift** (known as $E \times B$ drift) in the vertical direction
- ❖ Plasma rises, but then above some altitude, falls back down along field lines
- ❖ "Fountain effect"



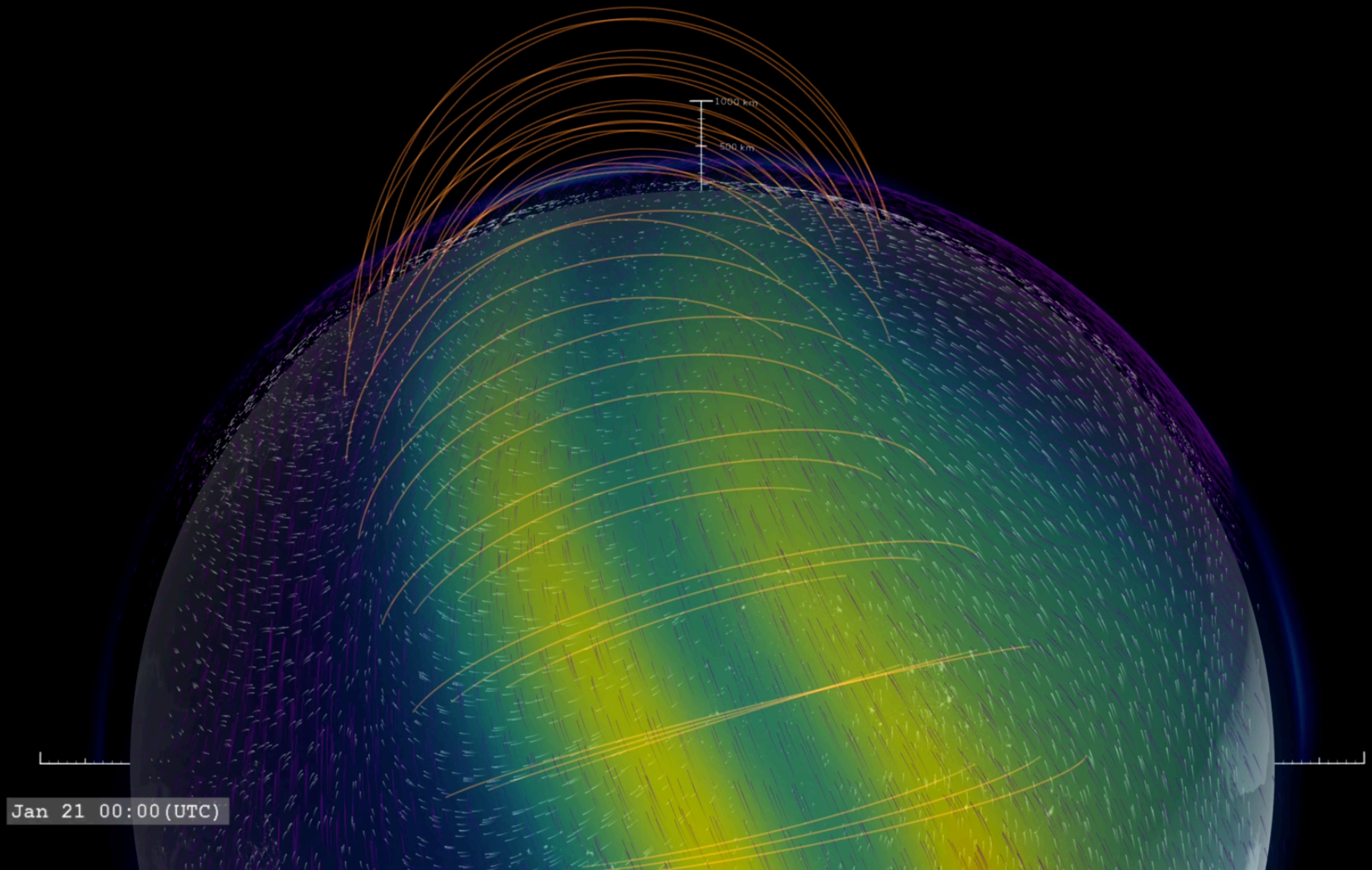
$$\vec{v}_d = \frac{\vec{F} \times \vec{B}}{q B^2}$$

$$\vec{F} = q \vec{E}$$

$$\vec{v}_d = \frac{\vec{E} \times \vec{B}}{B^2}$$

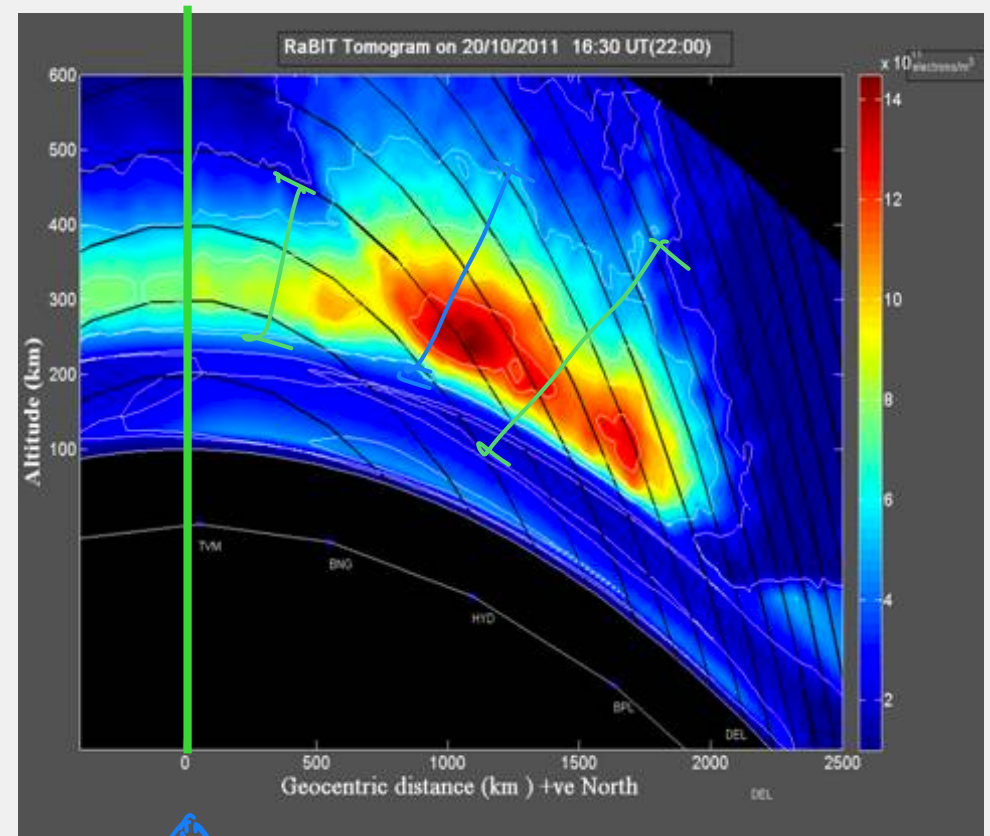
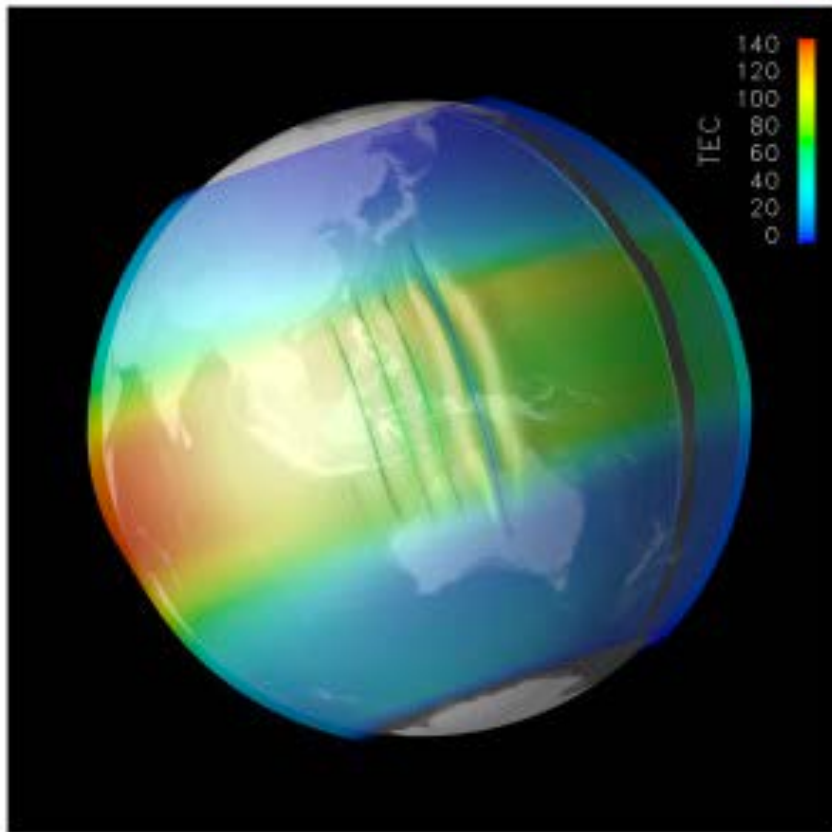
higher e^- density

Equatorial (ionization) Anomaly



Equatorial Spread-F

- ❖ Plasma instabilities that occur right after sunset cause the F-region to take on an array of structures
- ❖ Times scales from seconds to hours
- ❖ Spatial scales from cm to tens of km

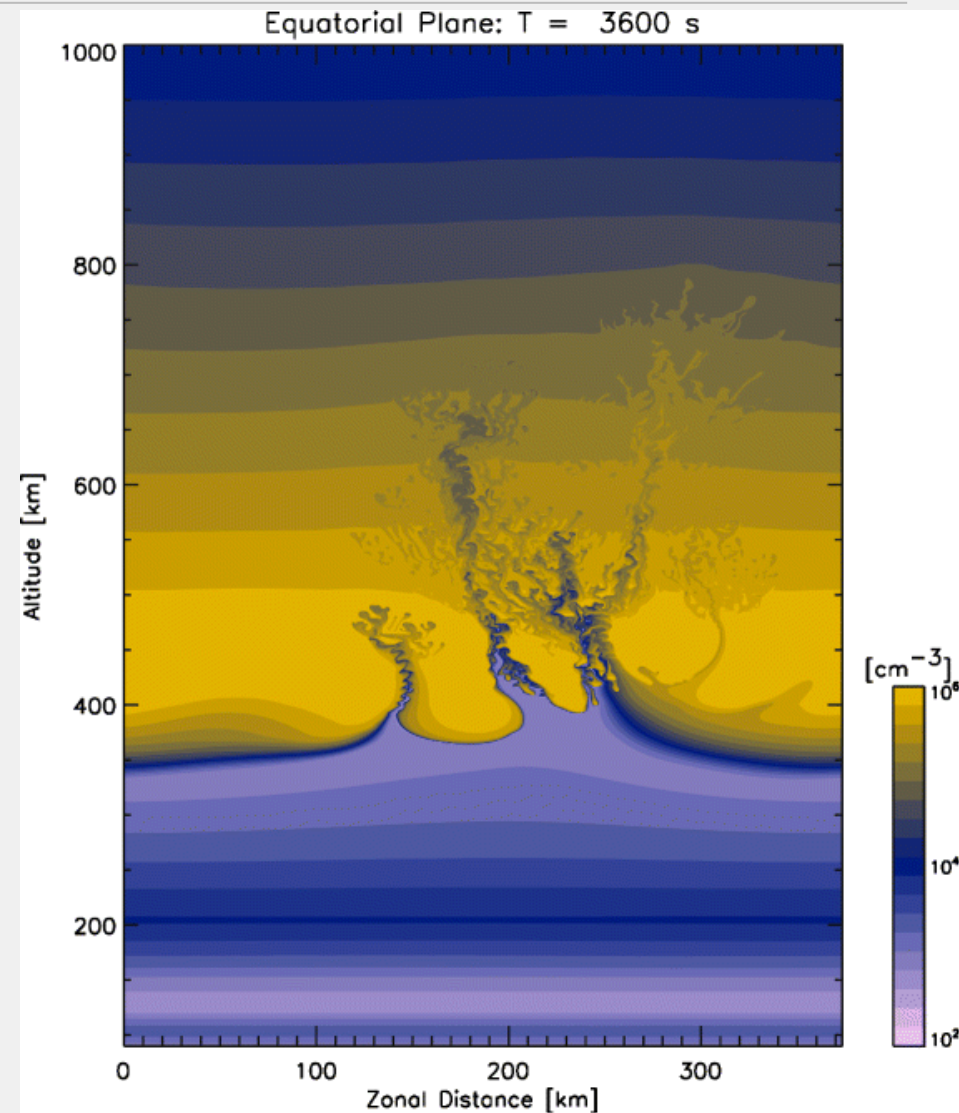
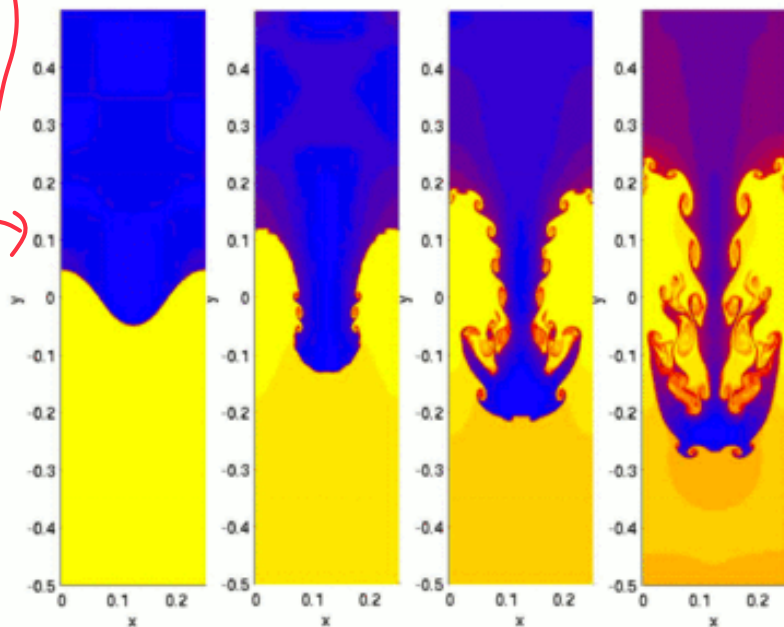


Equatorial Plasma Bubbles

- ❖ Instability due to heavier fluid on top of lighter fluid:

Rayleigh-Taylor Instability

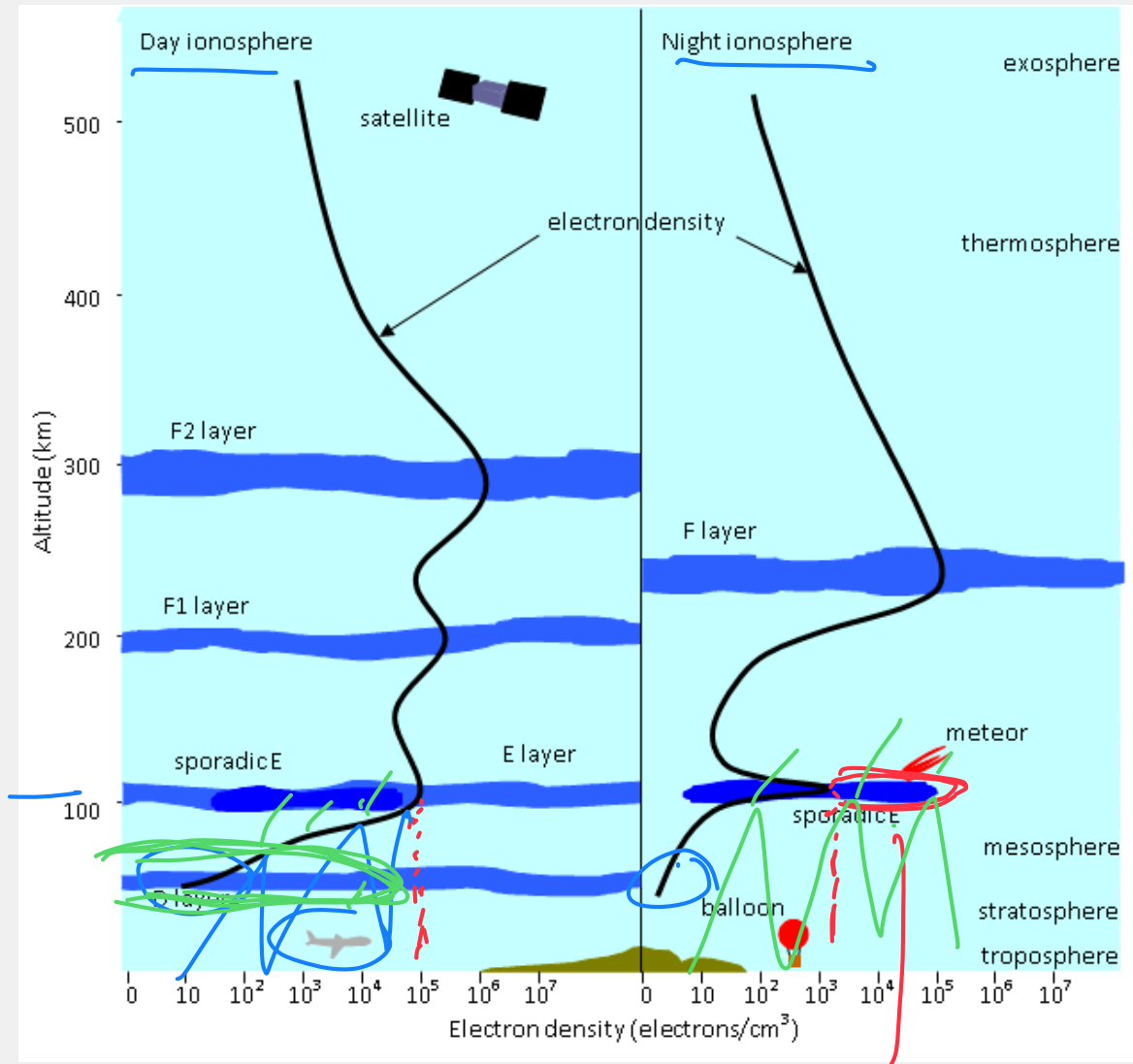
- ❖ Leads to rising bubbles with detailed structure and wide range of spatial scales, and large density variation



Kevin-Helmholtz instability

Mid-latitudes: Sporadic E (leading to higher e density)

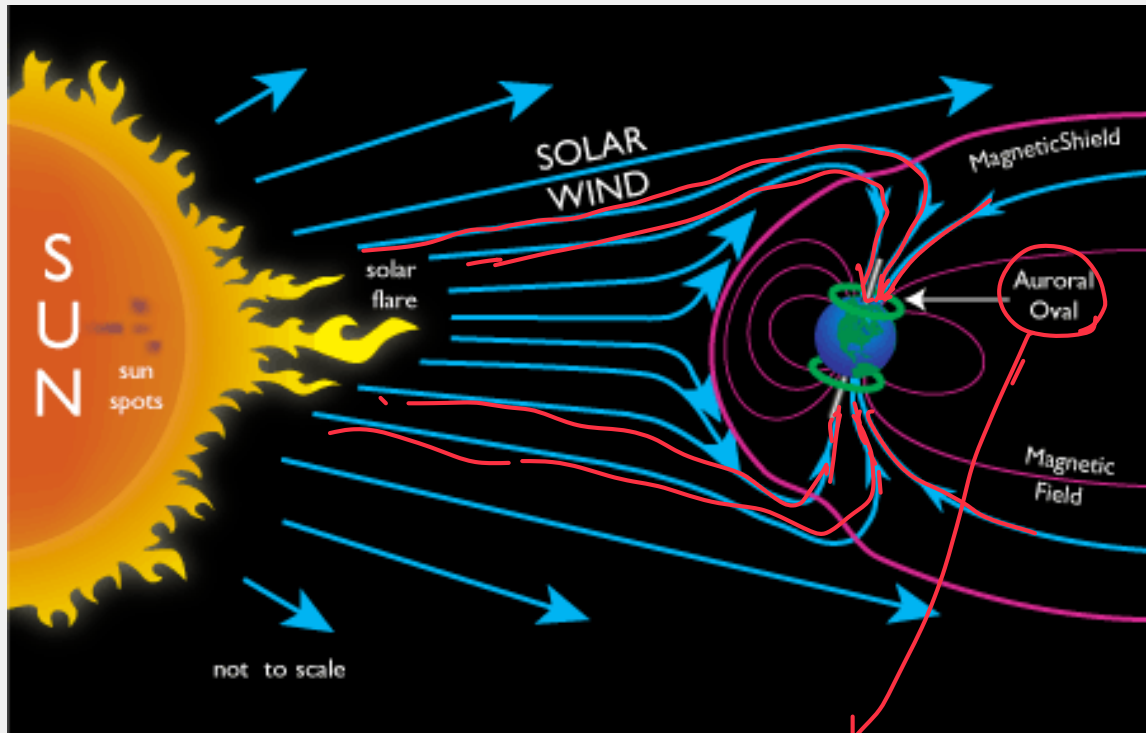
- ❖ Sporadic increase in E-region electron density, primarily at night, by orders of magnitude
- ❖ "Patch" of increased ionization
- ❖ Still not clear what the spatial scale is
- ❖ Observed by radio wave scattering
- ❖ Effect is most likely due to shear winds driving metal ions (from meteors) into thin layer



High-Latitudes: Polar Cap Absorption

- ❖ Solar Energetic Protons (SEPs) deposit their energy in the D-region of the ionosphere
- ❖ Flow along open magnetic field lines
- ❖ Increased D-region density adds to absorption of radio waves (discussed a bit later)

70 km

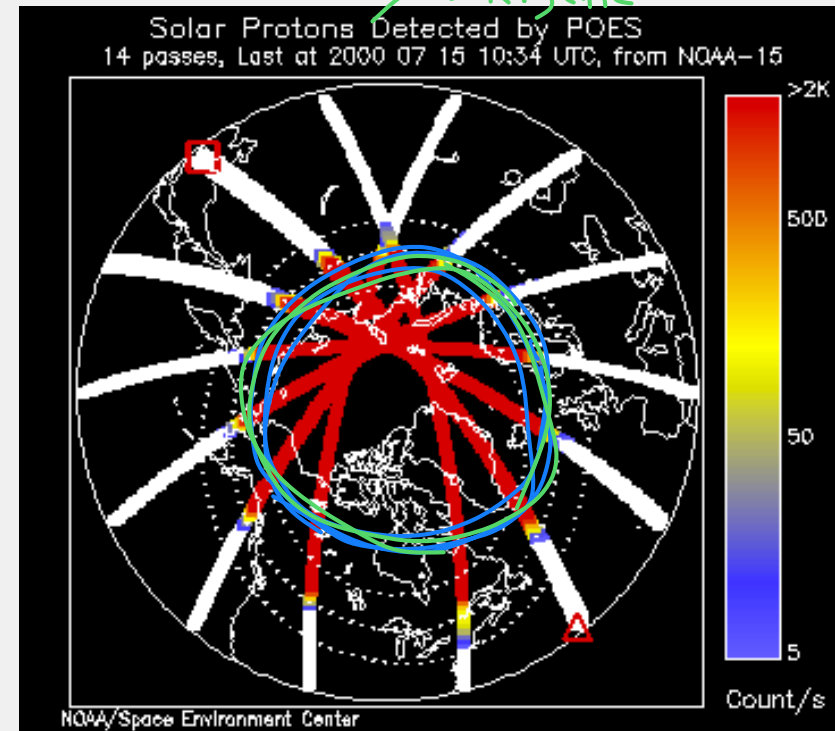


e^- higher

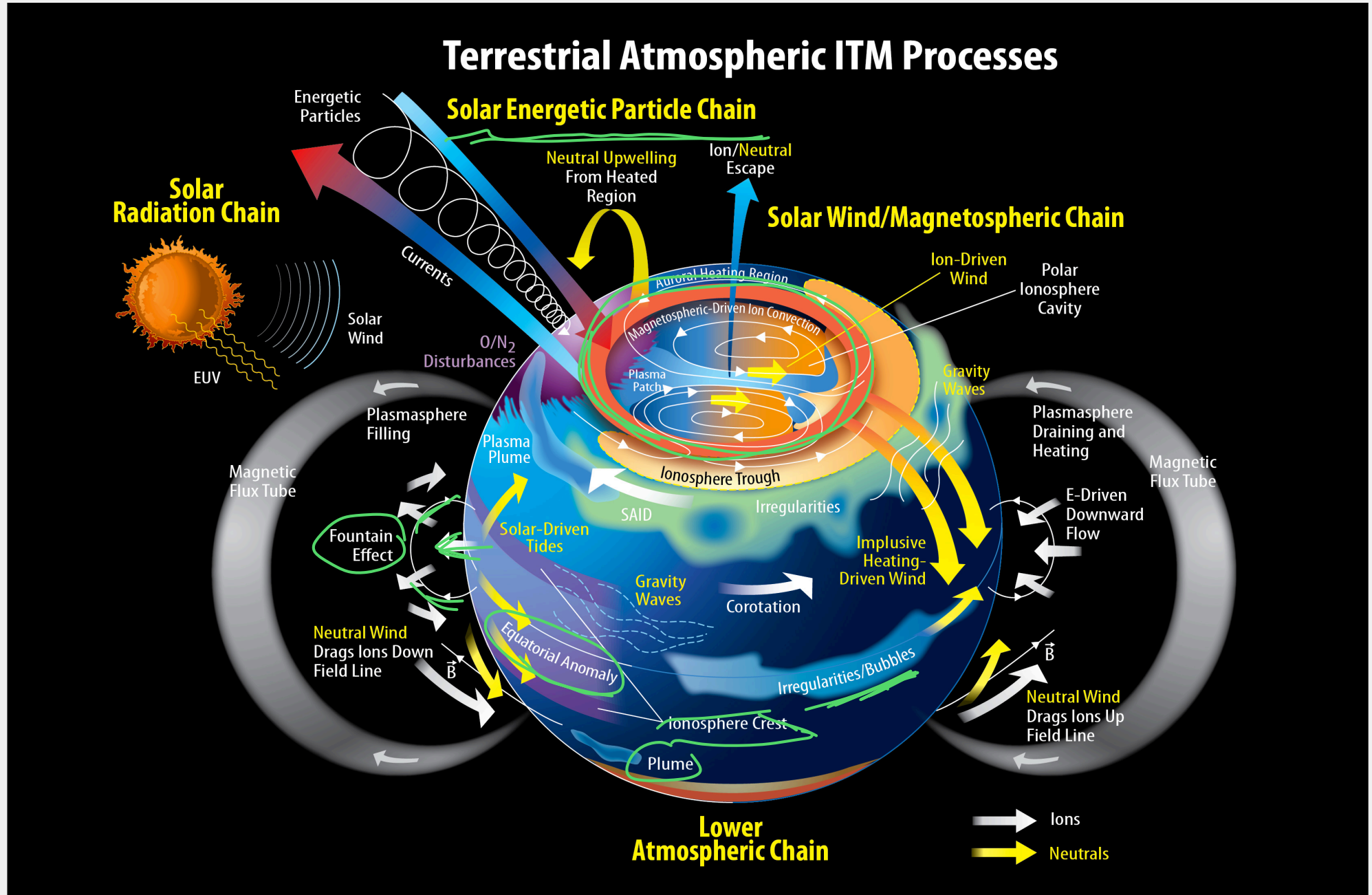
PCA

polar cap absorption

Solar Proton Energetic

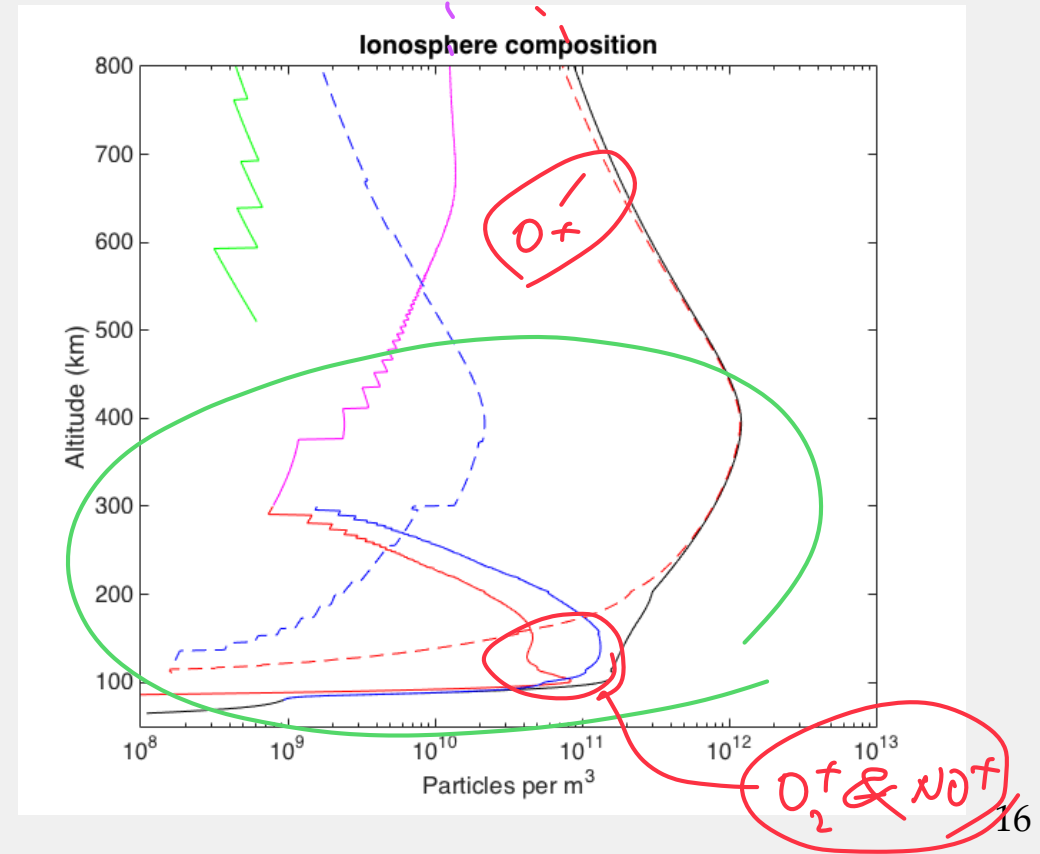
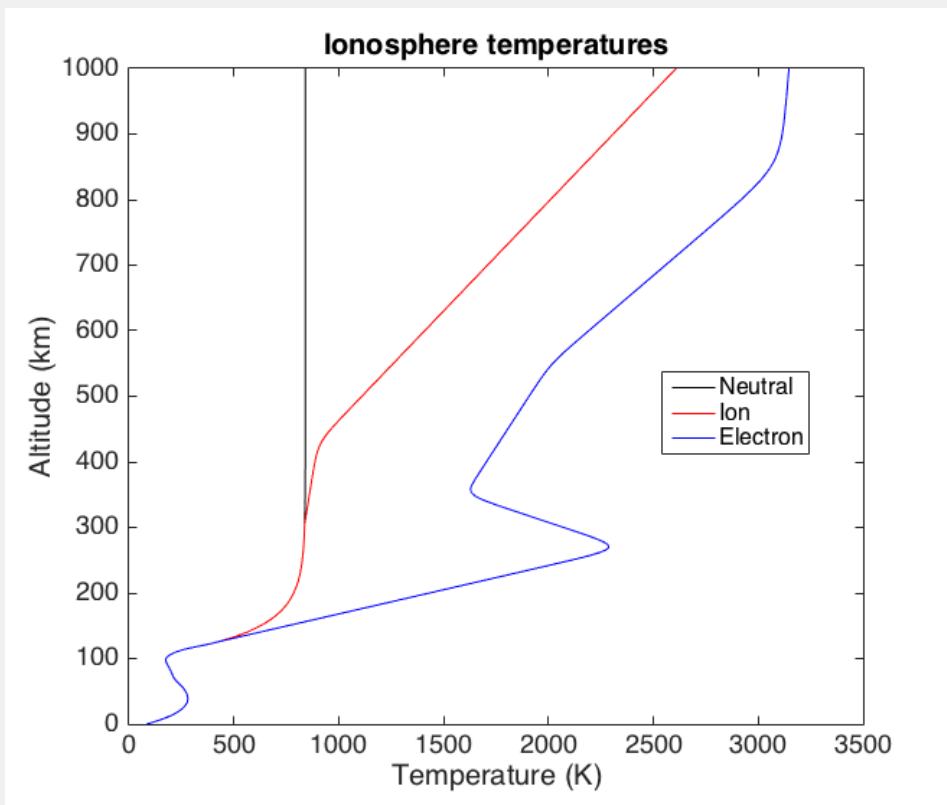


Lots more to the Ionosphere / Atmosphere system...



IRI model

- ❖ International Reference Ionosphere: IRI — HW3
- ❖ https://ccmc.gsfc.nasa.gov/modelweb/models/iri2012_vitmo.php
- ❖ IRI is ionosphere equivalent to MSIS
- ❖ Empirical model; major data sources are the worldwide network of ionosondes, powerful incoherent scatter radars (Jicamarca, Arecibo, Millstone Hill, Malvern, St. Santin), ISIS and Alouette topside sounders (spacecraft), in situ instruments flown on many satellites and rockets.
- ❖ IRI is updated yearly during special [IRI Workshops](#) at COSPAR and URSI meetings



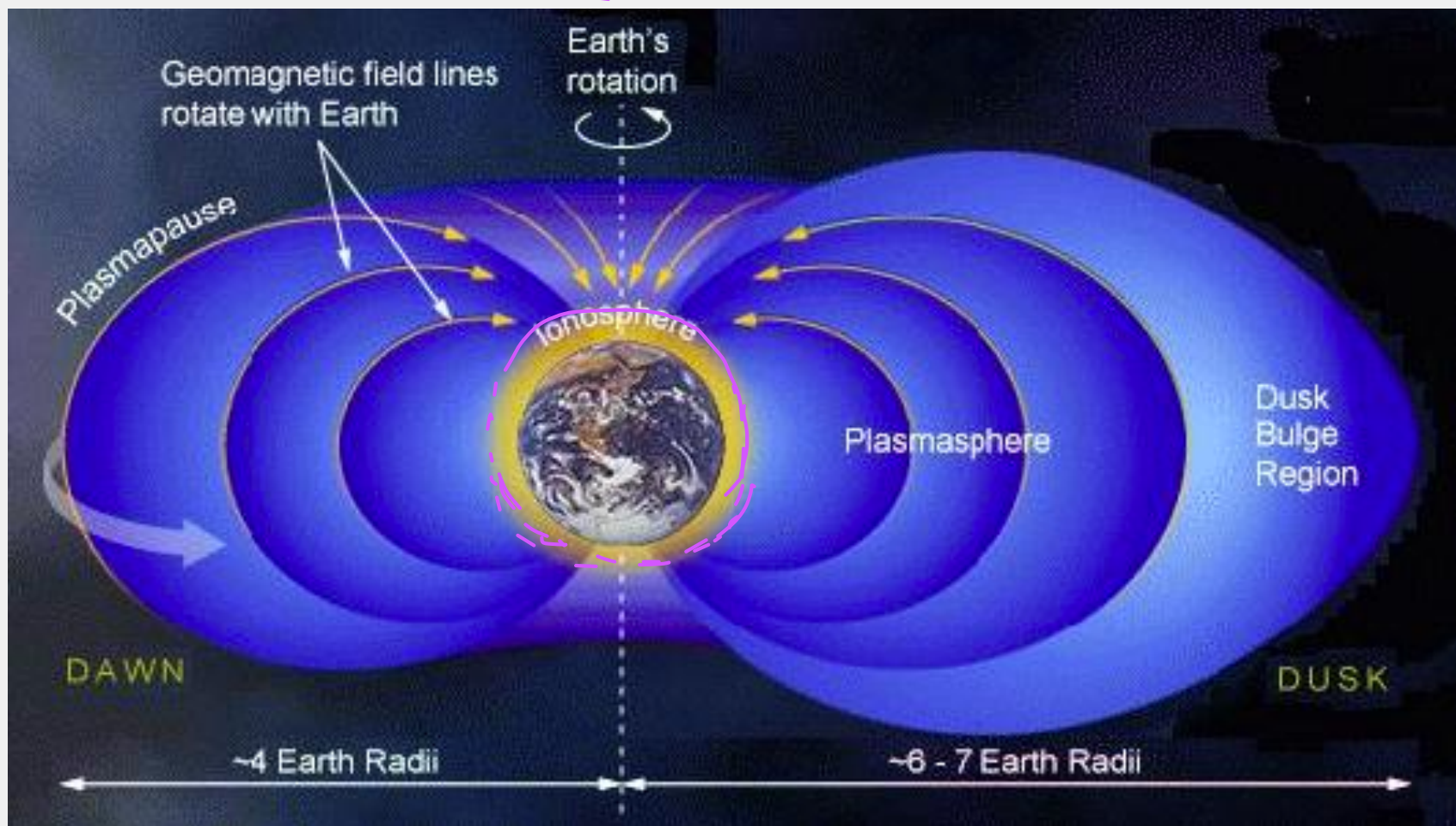


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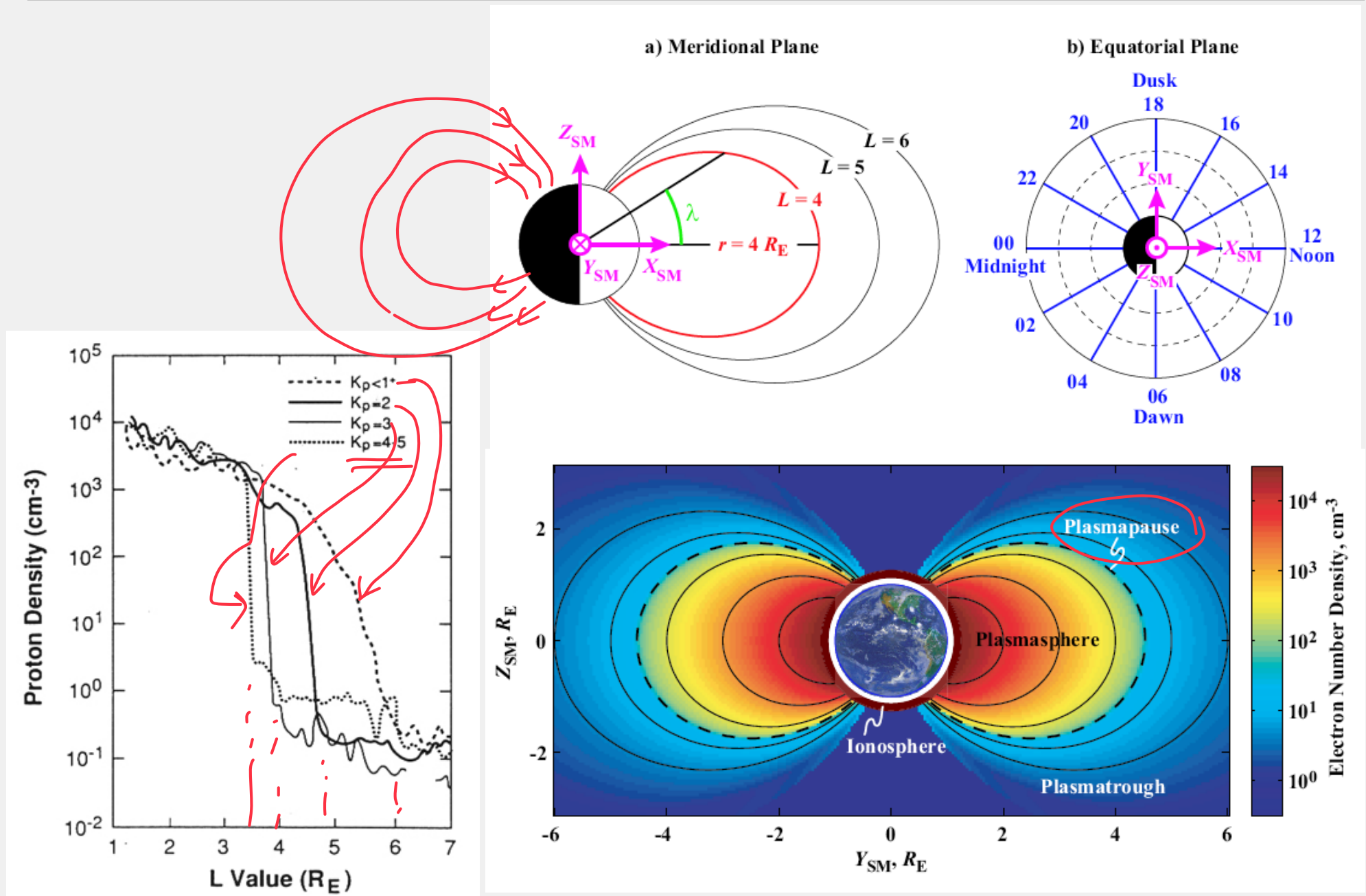
Plasmasphere

Plasmasphere

- ❖ Boundary between Toppedside ionosphere and Plasmasphere: where thermal pressure and magnetic pressure are equal
 - ❖ Plasma becomes confined by B-field
- ❖ Sometimes defined by altitude where H⁺ (protons) become dominant ion

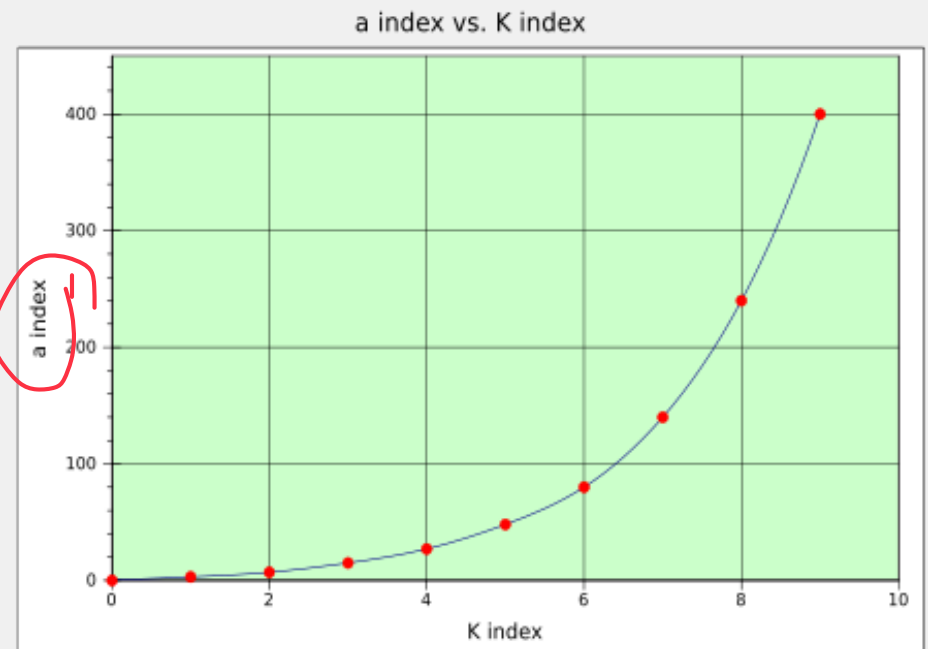
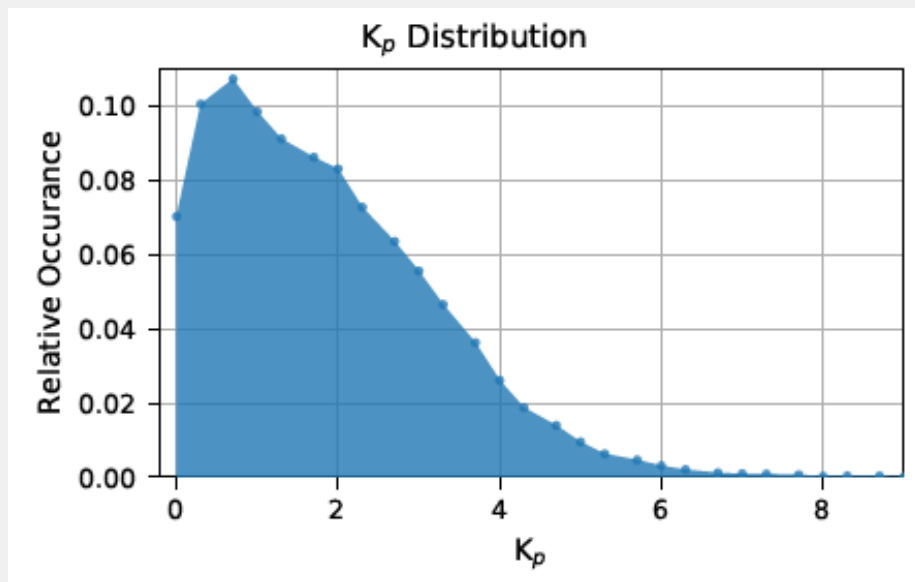


Plasmasphere structure and variation



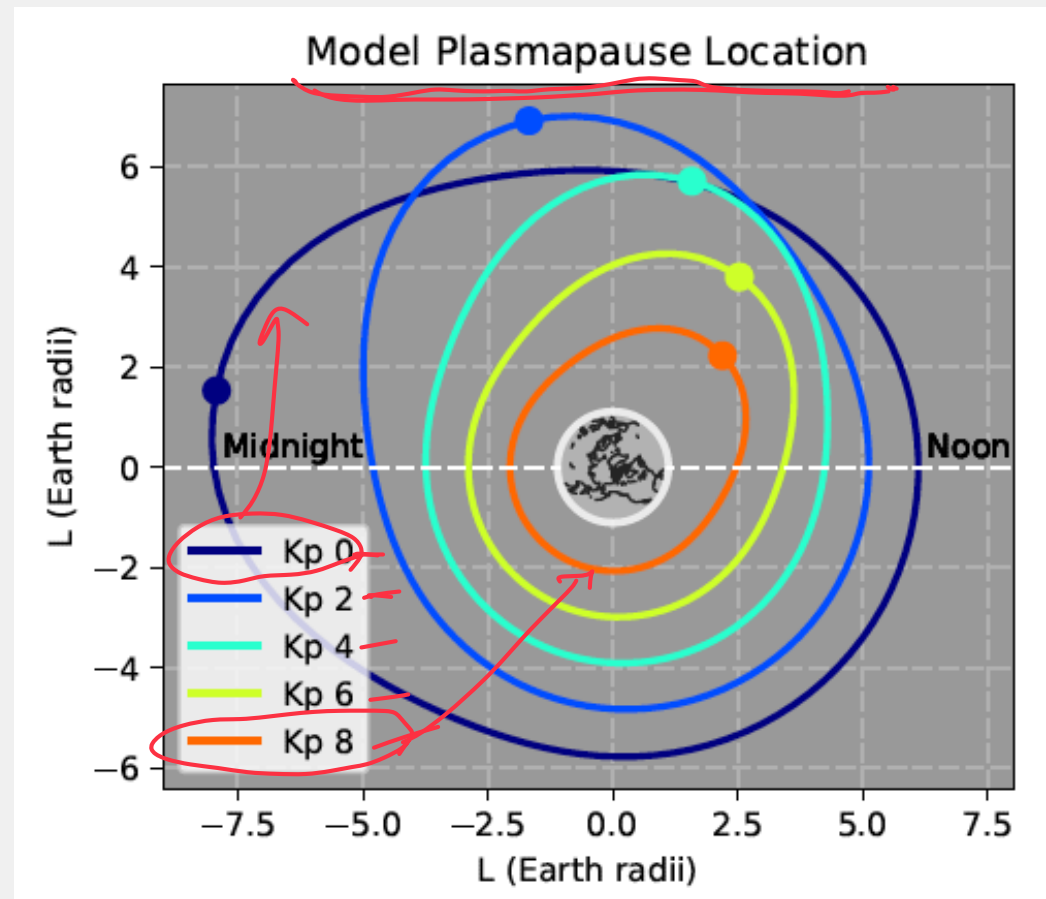
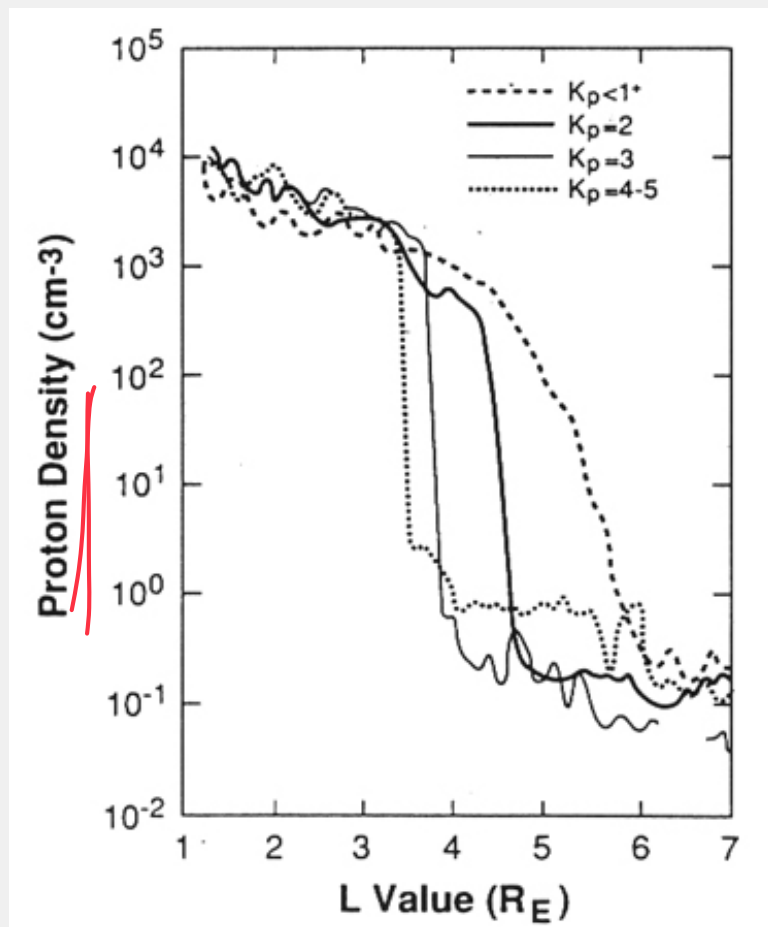
Kp index

- ❖ The **K-index** quantifies disturbances in the horizontal component of earth's magnetic field with an integer in the range 0-9
 - ❖ 1 = calm; 5 = geomagnetic storm; 9 = very, very rare
 - ❖ Measured at various ground magnetometer stations
 - ❖ Kp is average over all stations; "planetary" K index
 - ❖ quasi-logarithmic; Kp = 9 is about 10 times stronger than Kp = 3



Plasmasphere Gets "Squeezed"

- ❖ When K_p is high (in response to solar activity, of course), plasmasphere gets compressed
- ❖ Note "**plasmopause**" is where the density drops dramatically; used to be called Carpenter's Knee, after Don Carpenter of Stanford University, who passed away in 2019.



Plasmasphere Temperature

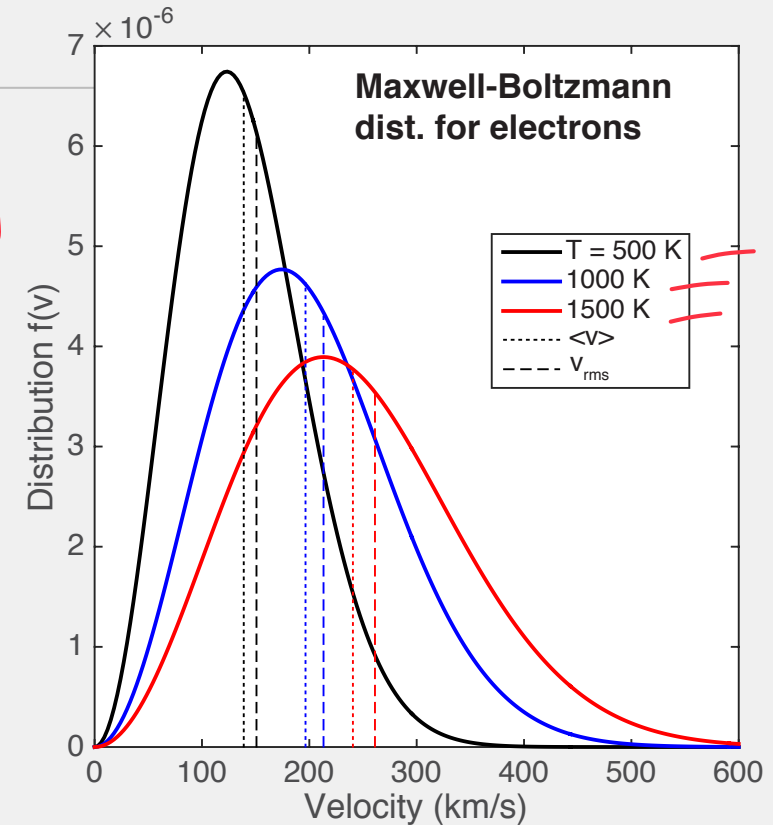
- ❖ The plasmasphere temperature is about 0.2–2 eV

$$f(v) = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 e^{-\frac{mv^2}{2kT}}$$

$k_B T$ unit of energy (J)

$$1 \text{ eV} \sim \frac{11600 \text{ K}}{1}$$

$$0.1 \text{ eV} \sim 1160 \text{ K}$$

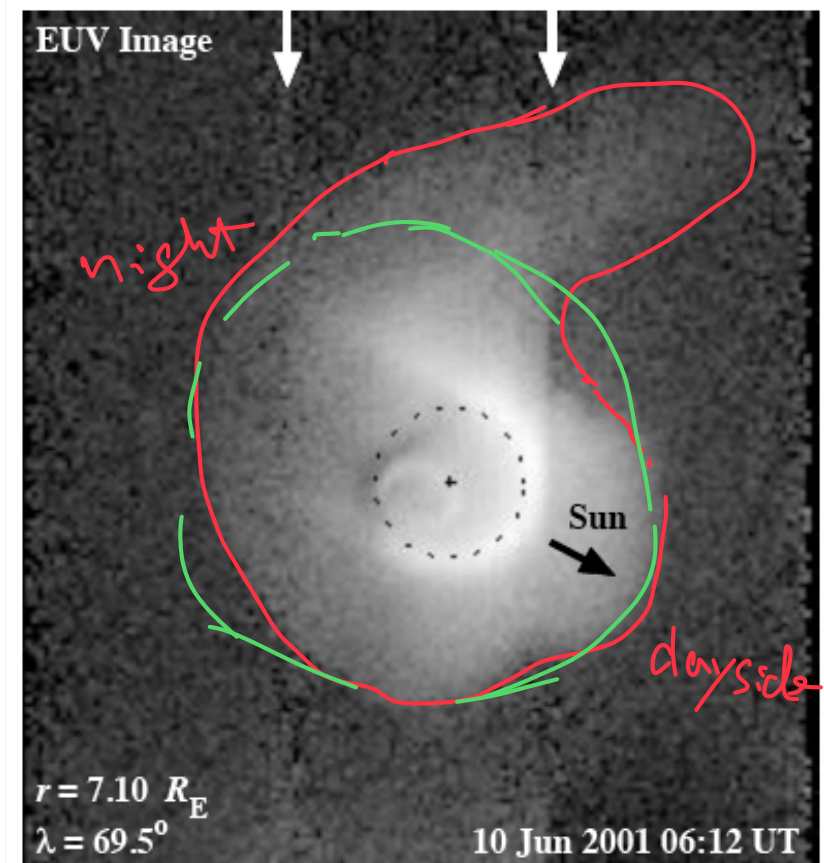
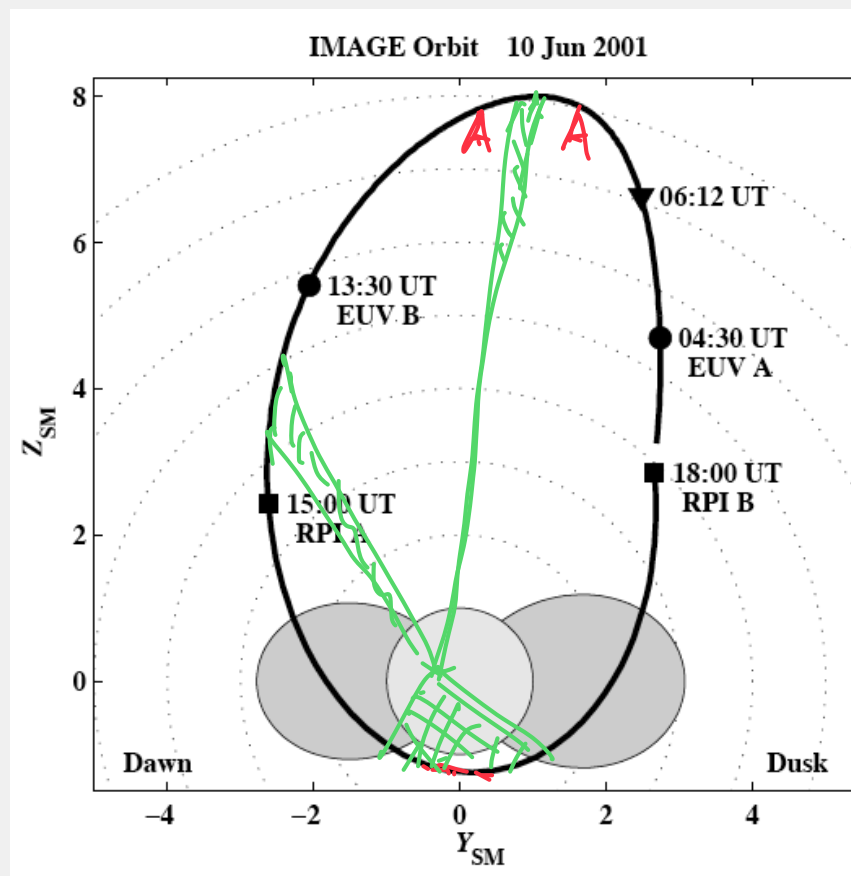


$$v_{th} \sim \sqrt{\frac{k_B T}{m}}$$

$$v_{rms} \sim \sqrt{\frac{3k_B T}{m}}$$

IMAGE-ing the Plasmasphere

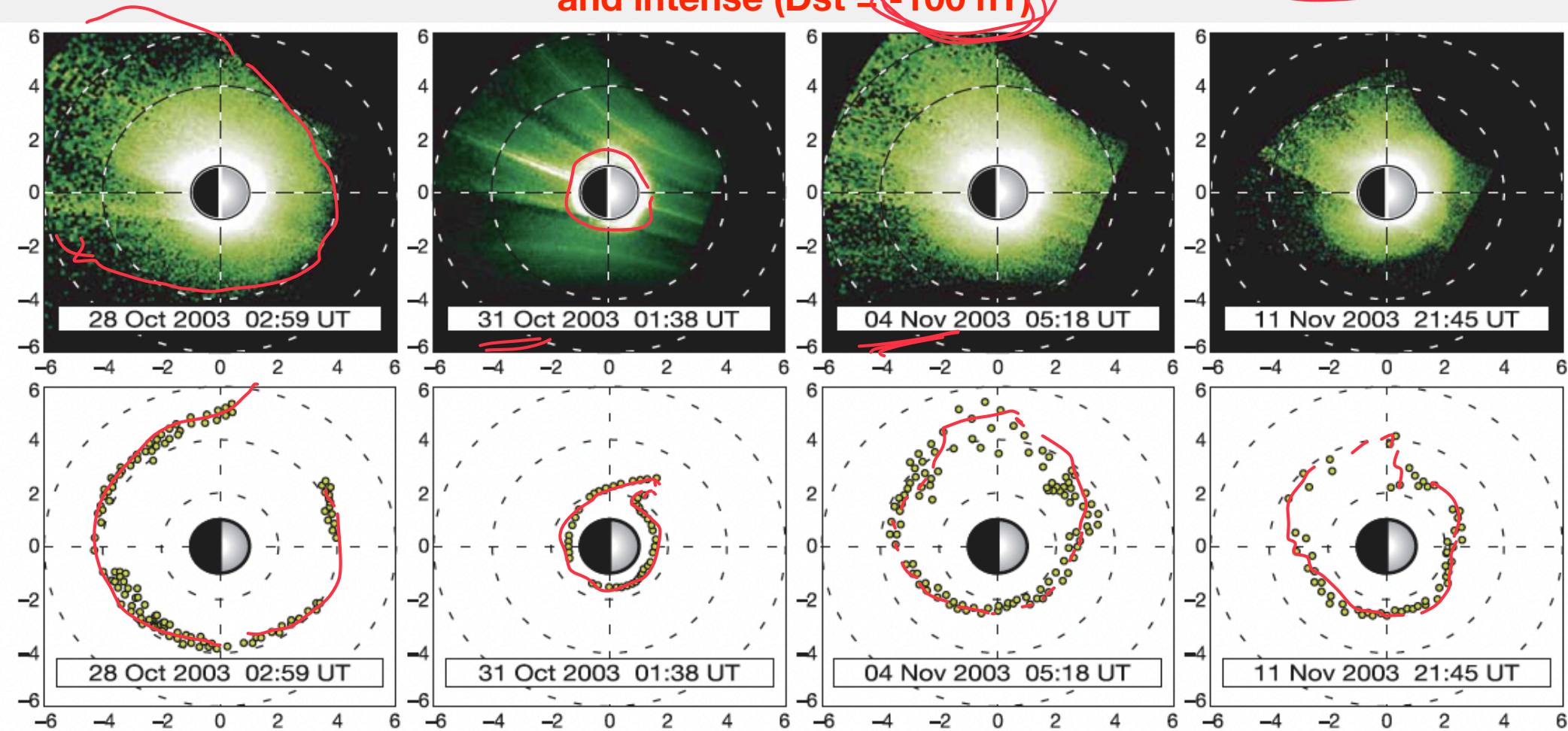
- ❖ H^+ makes up $>70\%$ of the plasmasphere ions, but has no visible emissions (it's just a proton...)
- ❖ He^+ constitutes the other 3–30% of the plasmasphere ions; resonantly scatters **30.4 nm** solar EUV radiation





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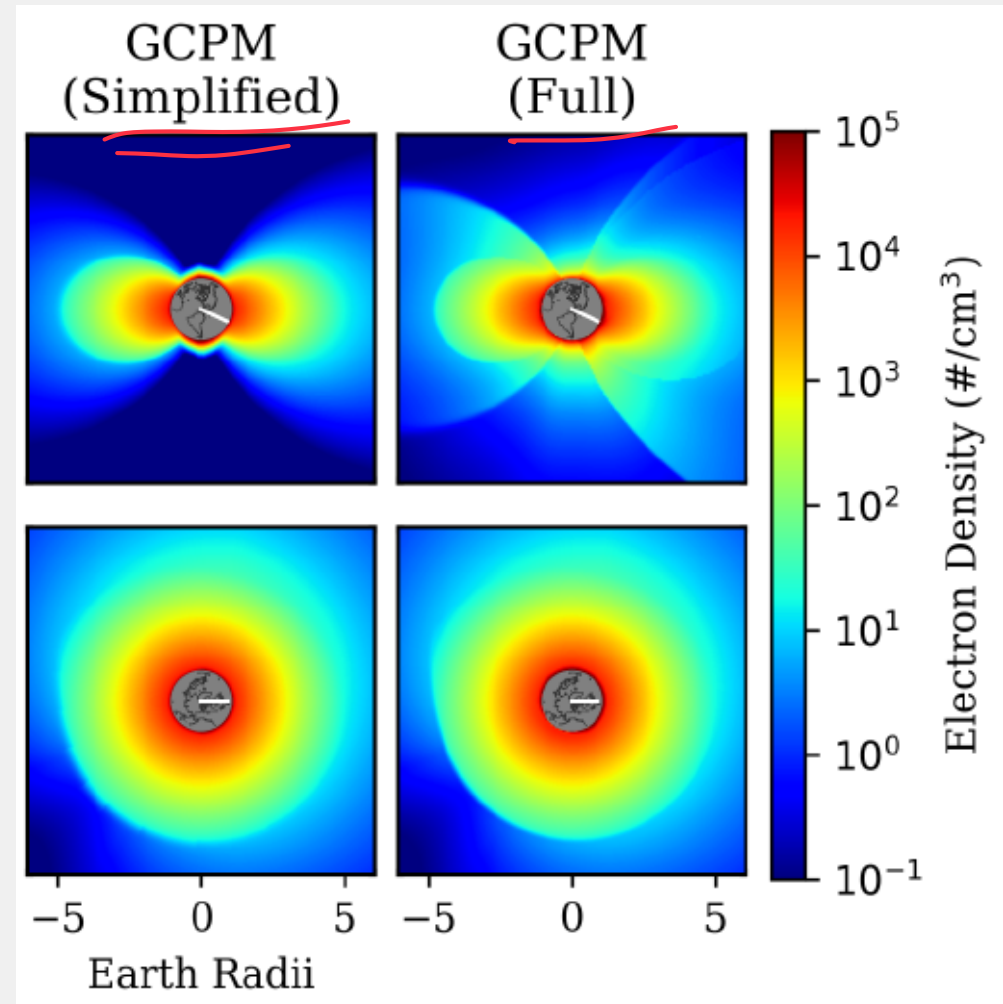
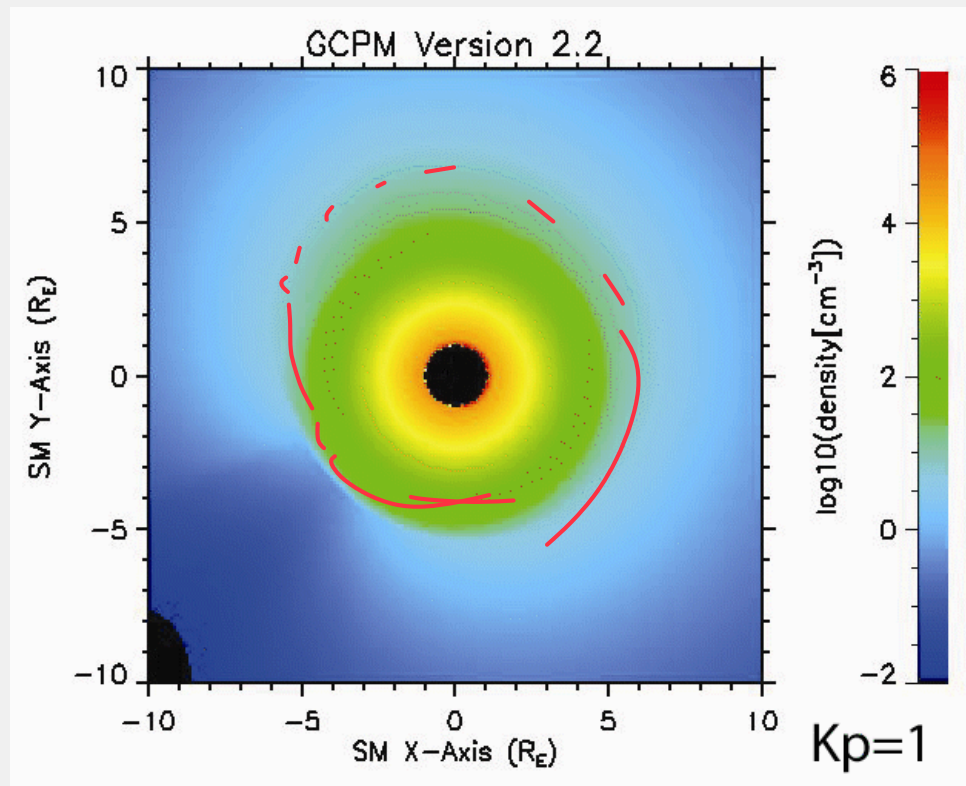
before and after two magnetic storms, one extreme intense ($Dst = -360$ nT) and intense ($Dst = -100$ nT)



(Baker, Kanekal, Li et al., Nature, 2004)

GCPM (Global Core Plasma Model) Plasmasphere Model

- ❖ Plasmasphere strongly affects spacecraft charging, as we'll see later, as well as radio wave propagation (below ~ 100 kHz)
- ❖ How do we model the density?
- ❖ Global Core Plasma Model (GCPM)





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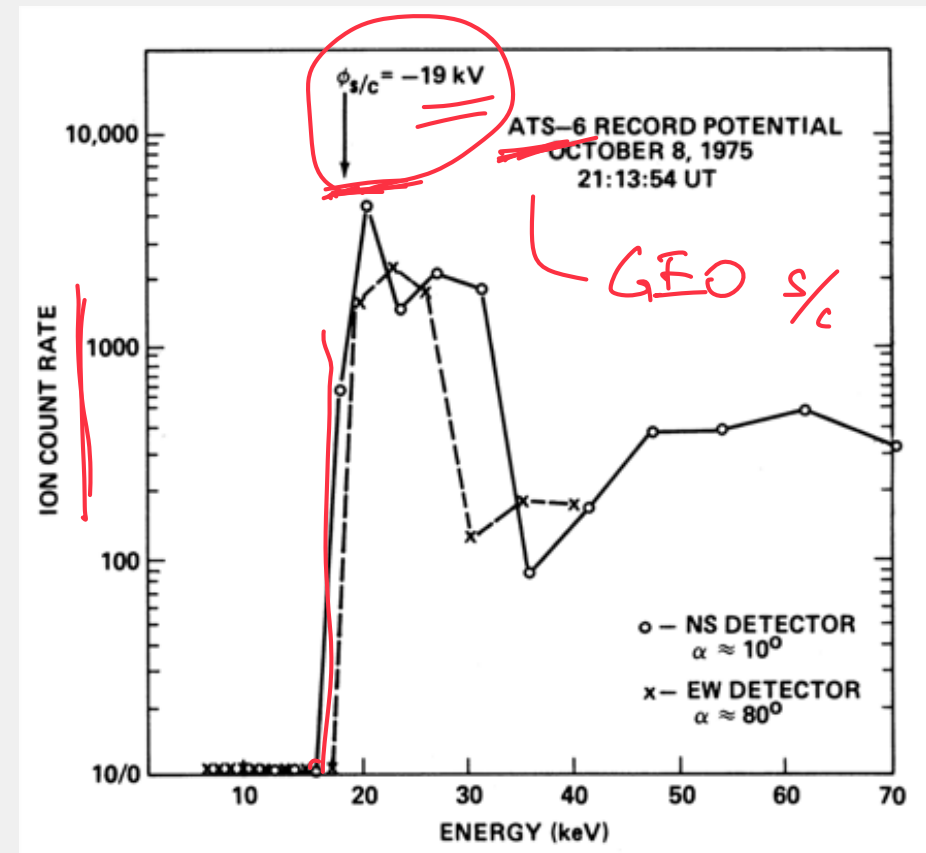
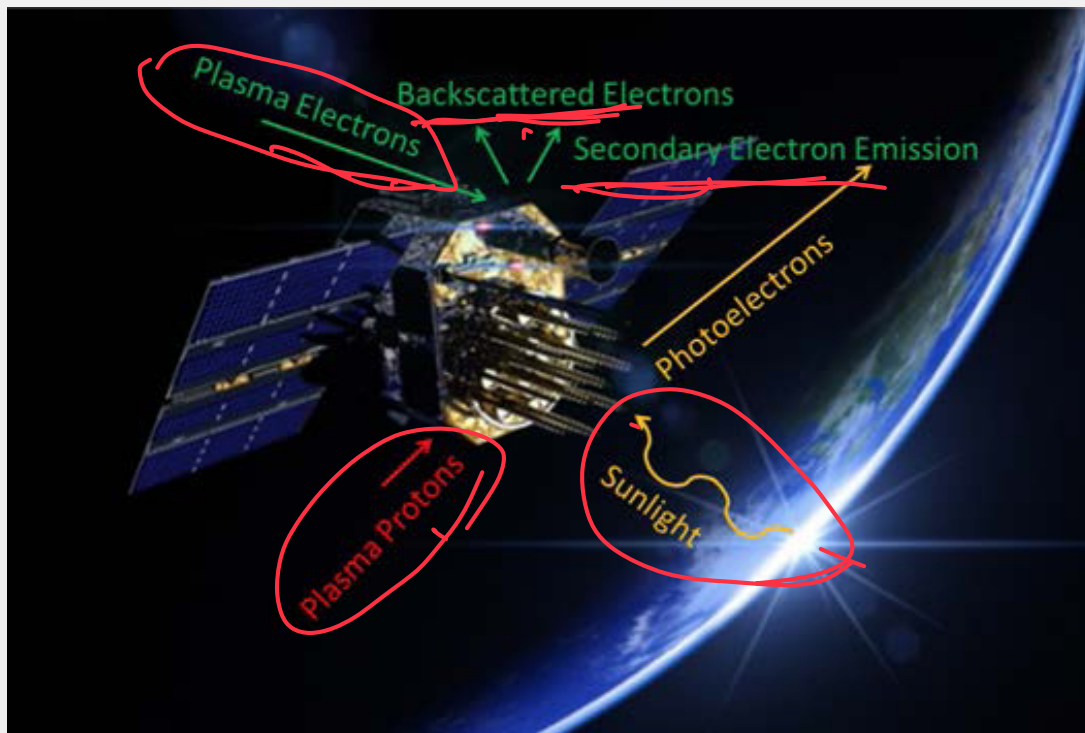
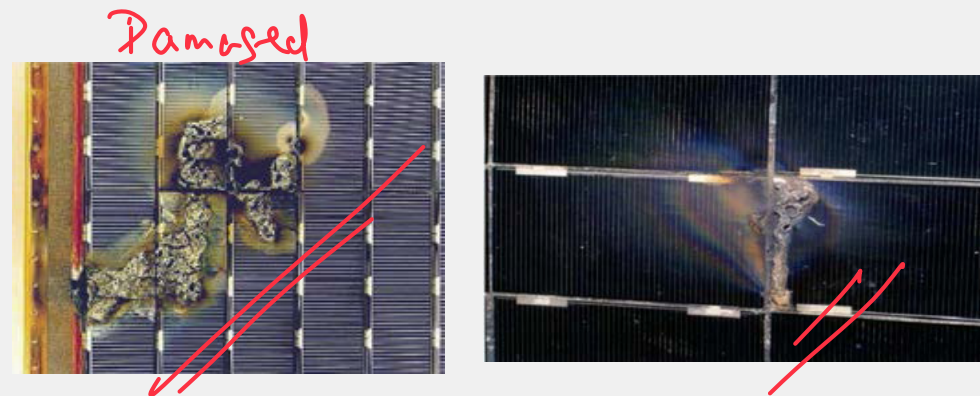
Ionosphere Effects

Ionosphere effects

- ❖ Spacecraft collect electrons (and ions) as they fly through the ionosphere (and similar regions of plasma), become negatively (or positively) charged: bad!
- ❖ Ionosphere affects propagation of radio waves: good and bad!
- ❖ **Others:**
 - ❖ Scintillations (rapid fluctuation of a radio signal's amplitude and/or phase)
 - ❖ Sputtering (again)
 - ❖ Solar array potentials

Spacecraft Charging

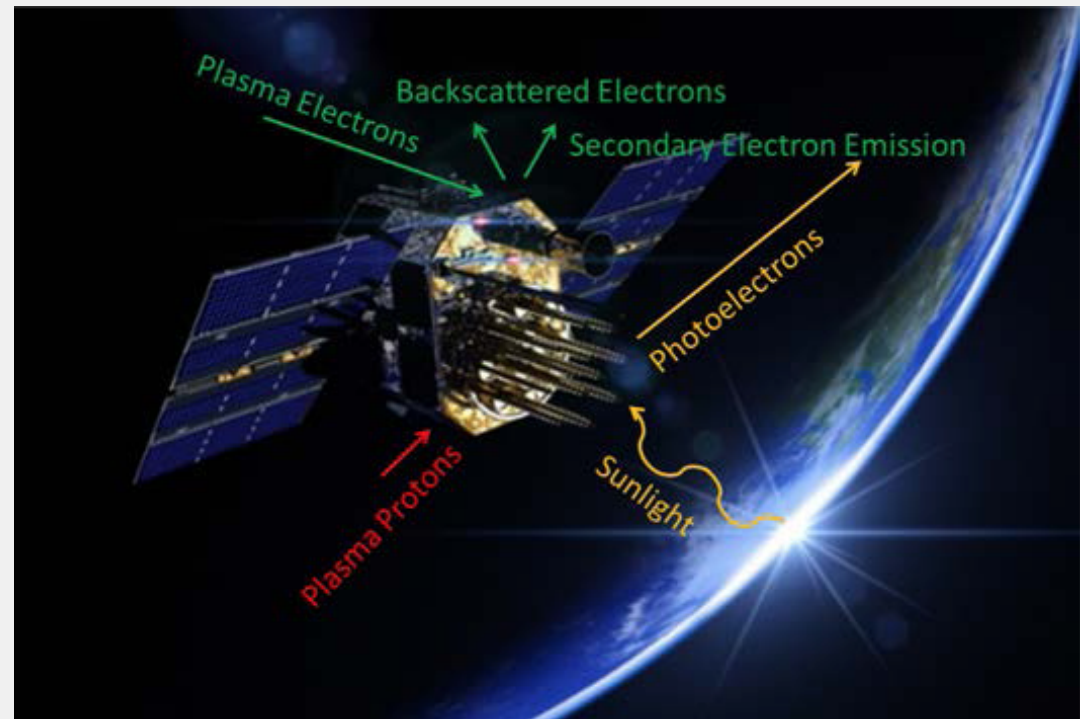
- ❖ Charge from the ionosphere / plasmasphere can accumulate on spacecraft surfaces
- ❖ Areas with different charge can lead to arcing or other discharge → damage!



Spacecraft Charging Equation

$$\frac{dQ}{dt} = \underline{I_e(V)} + \underline{I_i(V)} + \underline{I_{SEE_i}(V)} + \underline{I_b(V)} + \underline{I_{ph}(V)}$$

- ❖ Q = total charge on surface
- ❖ V = voltage of surface
- ❖ I_e = electron current from ionosphere
- ❖ I_i = ion current from ionosphere
- ❖ I_{SEE_i} = secondary electron emission due to ion current
- ❖ I_b = backscattered electron current
- ❖ I_{ph} = photoelectron current



What happens at equilibrium?

In LEO

$$\frac{dQ}{dt} = I_e(V) + I_i(V) + I_{SEE}(V) + I_b(V) + I_{ph}(V)$$

at night side

at 1000 K

$$I_e + I_i = 0$$

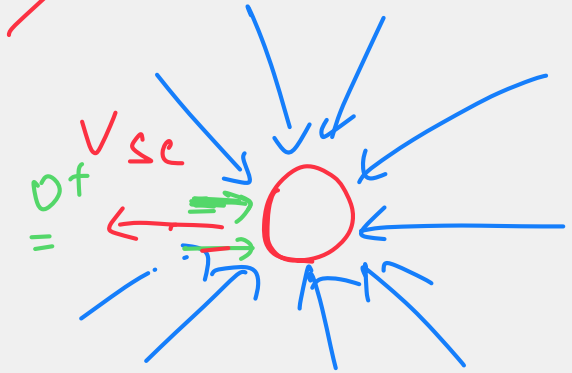
$$V_{sc} \sim 7.8 \text{ km/s}$$

$$V_{eth} \sim 200 \text{ km/s}$$

$$V_{ith} \sim 1.2 \text{ km/s}$$

$$V_{e,th} = \sqrt{\frac{3kT}{m_e}} = \frac{(3 \times 1.38 \times 10^{-23} \times 1000)^{1/2}}{9.11 \times 10^{-31}} = \frac{10^{-20}}{10^{-30}} = 10^{10} \text{ m/s}$$

$$V_{ith} : 200 \text{ km/s} / (16 \times 1836)^{1/2} \Rightarrow 1.2 \text{ km/s}$$



$$I_i = q_i n_i A_n V_{sc}$$

$$I_e = -\frac{1}{4} q_e n_e A_s V_{eth} e^{\frac{qV}{kT}}$$

$$= (4 \times 10^{10})^{1/2} = 2 \times 10^5 \text{ m/s}$$

$$= 200 \text{ km/s}$$

LEO Spacecraft charging

$I_i + I_e = 0$, $(n_i = n_e)$

$\Rightarrow V = \frac{kT_e}{q_e} \ln \left(\frac{4V_{sc}}{V_{eth}} \cdot \frac{A_n}{A_s} \right)$

Eq (23), in module note

πr^2

$4\pi r^2$

200 km/s

square (1u)

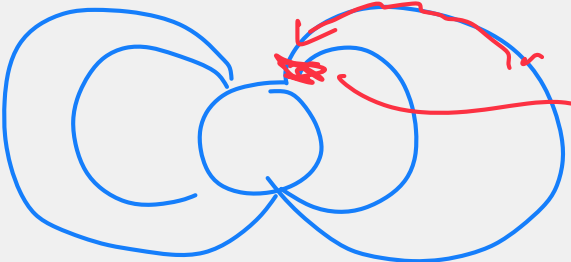
$A_n = a^2$

$A_s = 6a^2$

$V_{fl} = -0.27V$

size dependent? No

shape dependent? Yes



$T_e \sim 200 \text{ eV} \Rightarrow \frac{200 \times 11600}{2 \text{ MK}}$

GEO spacecraft charging

GEO: not in the ionosphere anymore

- ❖ Which space environment is relevant?

$$T_e \sim 1000 \text{ eV} \sim T_e \sim 10^7 \text{ K}$$

$$V_{sc} \sim \sqrt{\frac{GM}{r}} \sim 3 \text{ km/s}$$

$$V_{e+th} = 10,000 \text{ km/s}$$

$$V_{i+th} \sim \underline{460 \text{ km/s}} \quad (p^+ : T_p > 4T_e)$$

$$I_{ph} = 0$$

$$I_e = -\frac{1}{4} q_e n_e v_{e,th} A_e \left(e^{q_e V / k_B T_e} \right)$$

Eq (21)

$$I_i = \frac{1}{4} q_e n_0 v_{i,th} A_i \left(1 - \frac{q_e V}{k_B T_i} \right)$$

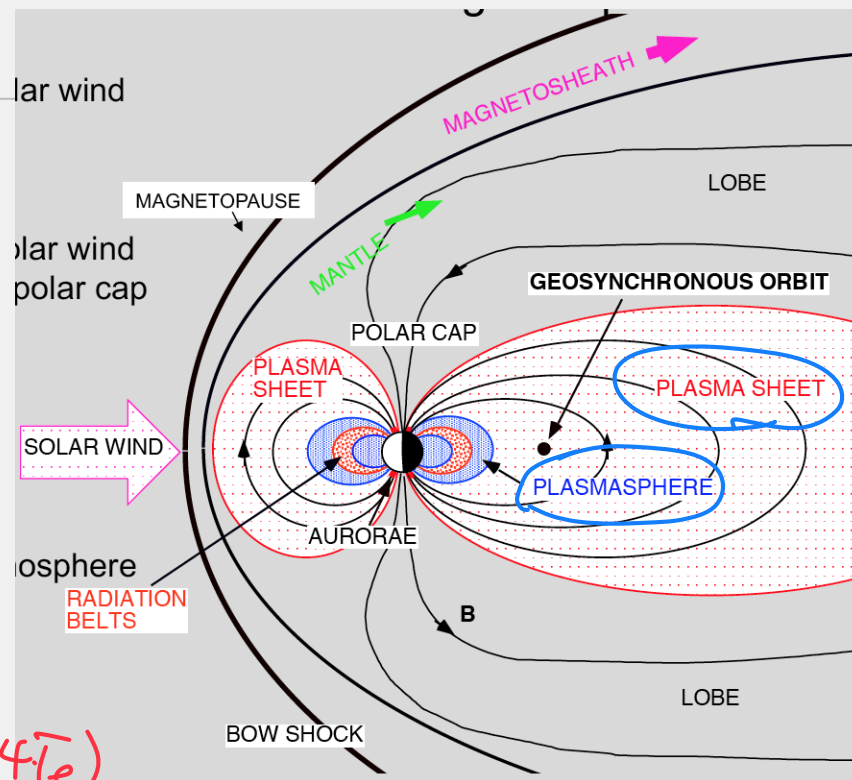
Eq (25)

$I_e + I_i$



$$V_{fl} = \frac{k_B T_e}{q_e} \ln \left[\sqrt{\frac{m_e}{m_i}} \left(1 - \frac{q_e V}{k_B T_i} \right) \right]$$

Eq (26)



$$V_A = -2500 \text{ V}$$

GEO spacecraft charging

$$I_e = -\frac{1}{4}q_e n_e v_{e,th} A_e \left(1 + \frac{q_e V}{k_B T_e} \right) \quad \text{for } V > 0$$

$$I_i = \frac{1}{4}q_e n_0 v_{i,th} A_i \left(e^{-q_e V/k_B T_i} \right) \quad \text{for } V > 0$$

What about the photocurrent?

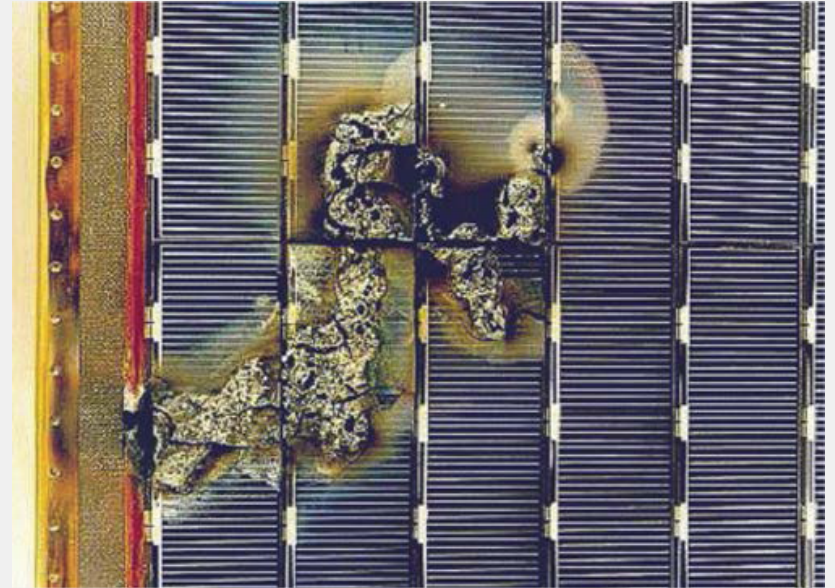
- ❖ Photocurrent is electrons leaving the spacecraft surface due to photoelectric effect from solar UV
- ❖ Electrons going **away from** spacecraft = current **towards** spacecraft

Eq (28)

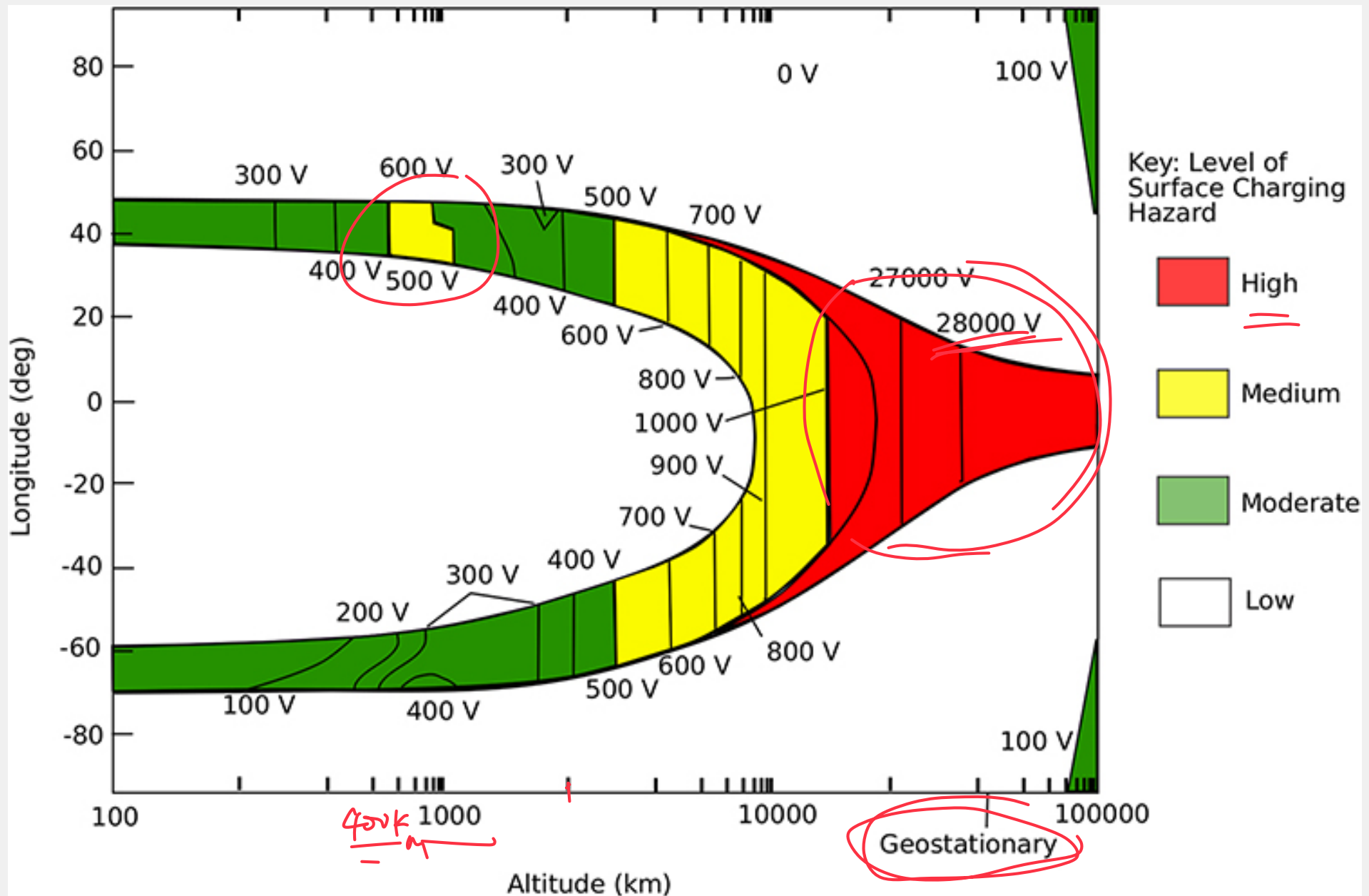
Differential Charging

Effects of Spacecraft Charging

- ❖ We rarely care about the absolute potential: simply relative to “space”
- ❖ Effects occur when there is significant potential difference (i.e. voltage) between two regions of spacecraft
- ❖ **Electrostatic Discharge:** Transfer of charge through spacecraft material due to electric field
 - ❖ Starts at -150 to -500 V
 - ❖ Produces electromagnetic pulse / interference
- ❖ **Dielectric Breakdown:** materials break down at some electric field (air: 3 MV/m at sea level)



Where is spacecraft charging a concern?



LEO Discharge events occur over South America

- ❖ And at high Latitudes (~ 70 degrees).
- ❖ **Why?**

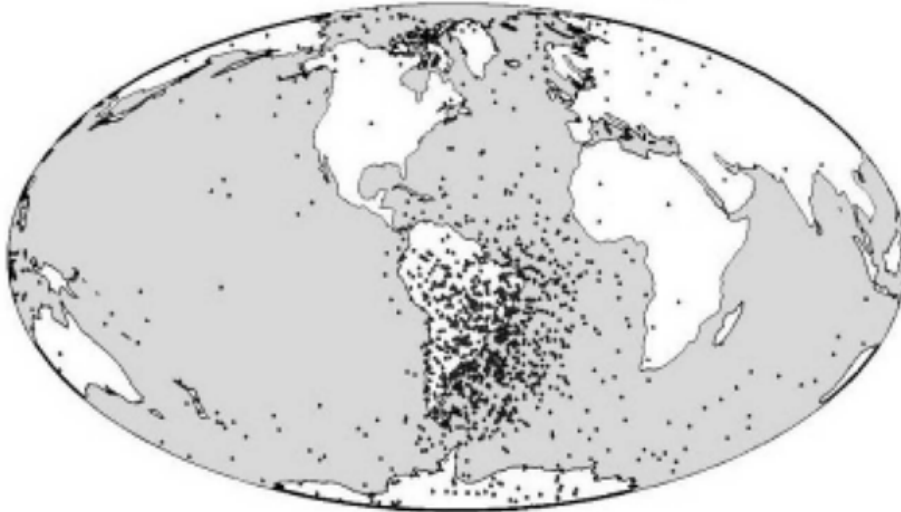


Figure 5. Single-event upsets on the low-altitude UoSAT-2 amateur-radio communications satellite. The majority, from protons in the inner radiation belt, occur in the South Atlantic Anomaly over South America and the South Atlantic Ocean. Upsets scattered over the rest of the globe are from galactic cosmic rays [6] (courtesy of NOAA).

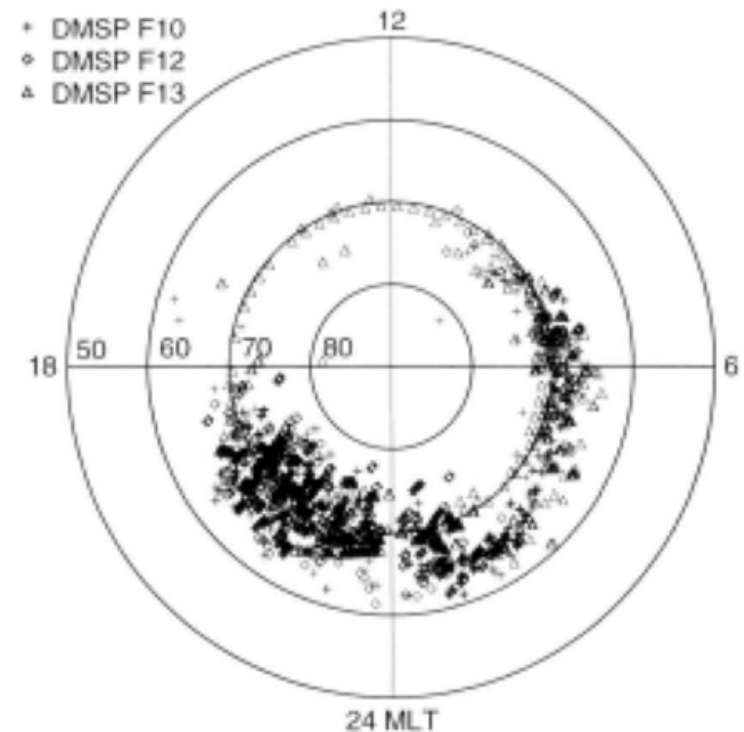


Figure 8. The location in magnetic latitude and magnetic local time of surface-charging events measured on the DMSP satellites at low altitude in the auroral zones [22] (courtesy of the Air Force Research Laboratory).

GEO: Discharge events occur in the post-midnight and early morning...

❖ Why?

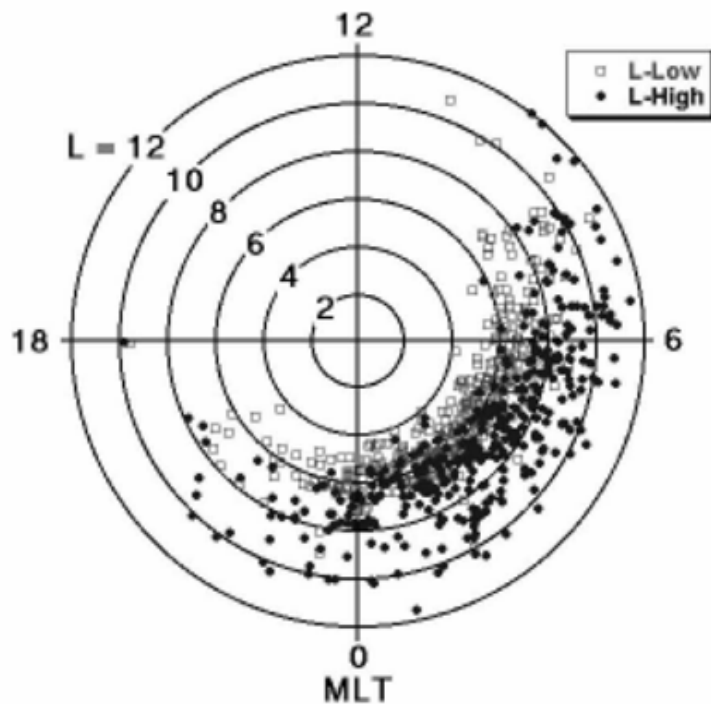
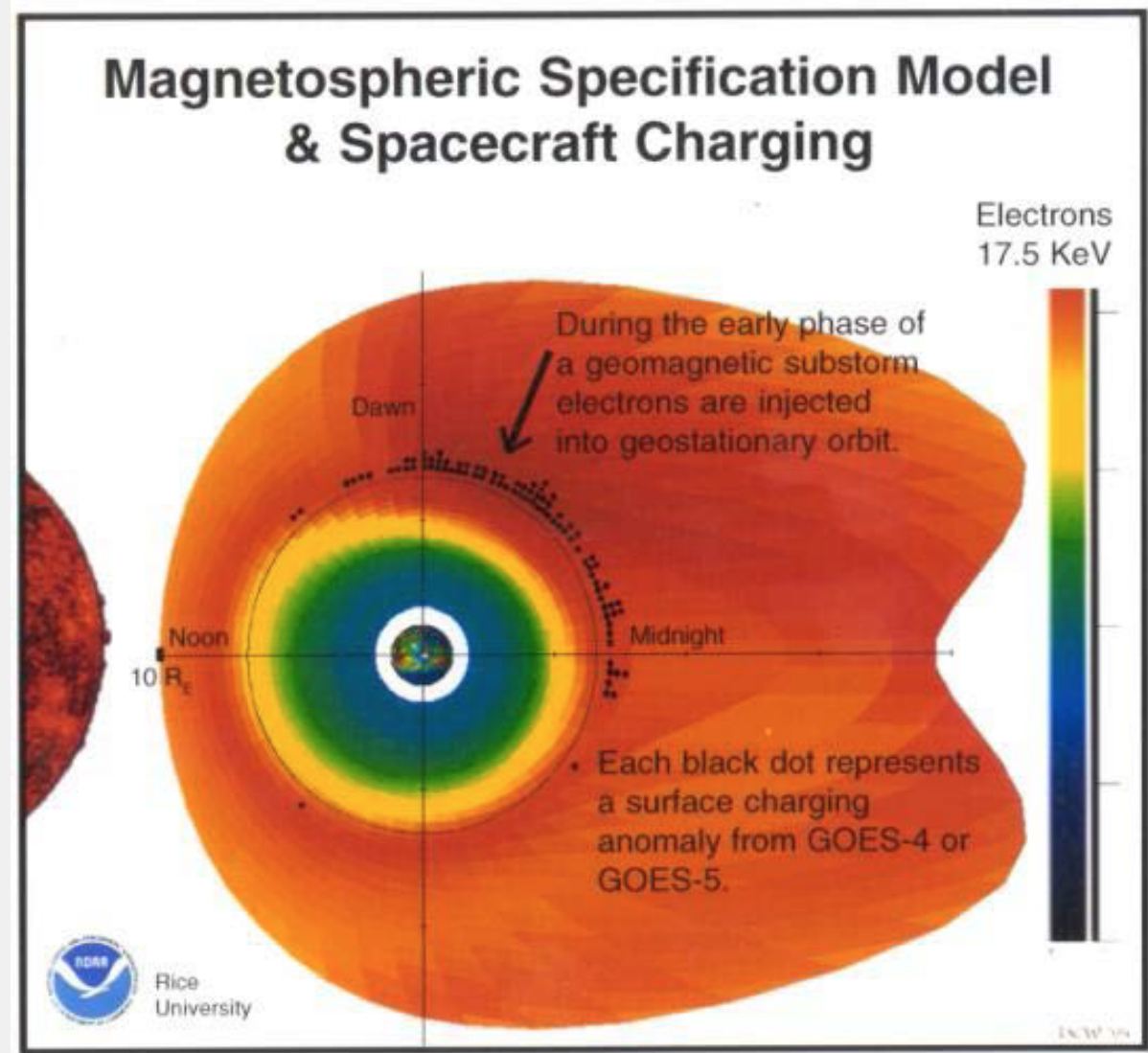


Figure 10. The location in Magnetic Local Time and L-shell where the spacecraft frame potential was less than -30 V in a Molniya orbit. The squares indicate the low L-shell limit of the charging (no data were taken below $L \sim 4$) and the circles show the high L-shell limit for each orbit. The points were not connected to simplify the drawing.



Radio Wave Propagation

- ❖ Time for some plasma physics!
 - ❖ Plasma oscillations
 - ❖ Plasma frequency
 - ❖ index of refraction (from Maxwell's equations)
 - ❖ Add collisions: absorption
 - ❖ MUF, LUF, X-ray effect